

Edition: MONARCH integrated machine T19.0

ACCORDING TO STANDARD: GB 16899-2011

User manual of escalator and moving walk

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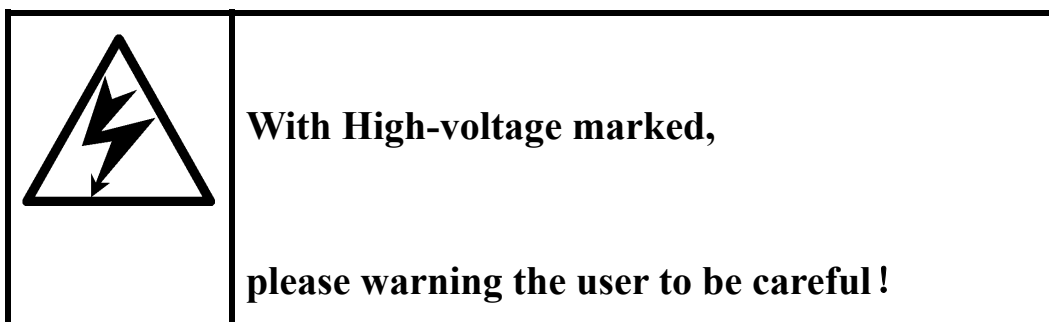
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Note: The manual will change according to the products and will not inform.

1 Precautions

1.1 Safety and precautions



This equipment's voltage will cause serious electric shock and burns, even cause death. When you use or go close this equipment, please be high degree of vigilance.

This equipment design for monitoring the safety functions of the escalator and moving walk. If the installation is not correct, the escalator and moving walk will have safety risks. This equipment's control power is DC24V switch power. Improper operation will hurt human body.

The people for design/install/commission and maintain this system must be training and have related experience. They must read this documents and other user manual carefully and save them for use them in the future. This document and other user manual should be sending to the user.

We cannot make sure the safety of the people according to the stop/start or input signal, they can not isolation the dangerous voltage and the output of this equipment. Before connect the wires, must separate the escalator or moving walk by the insulation elements in line with national standards.

1.2 Environment of using

The transport/store and using of the escalator and moving walk must according to the environment requirements of this documents and related user manual. This equipment can not bearing excessive force, can avoid the damage.

The electric parts and equipment on escalator and moving walk should have good ventilation, the requirement of the using environment according to the follow:

——working temperature: $-5^{\circ}\text{C} \sim +60^{\circ}\text{C}$

——working humidity: 25%RH $\sim$ 85%RH

——store temperature: $-10^{\circ}\text{C} \sim +60^{\circ}\text{C}$

——store humidity: 10%RH $\sim$ 85%RH

The escalator should also far away from the conductive material, corrosive gases, flammable gas, metal powder, oil mist, dust and so on.

1.3 Compliance and regulations

——The people of installation should according to all related provision, for example: the wiring/accident defense and electromagnetic compatibility provision.

—The safety device designed related in this document is according to the safety provision of national standard GB16899-2011 and GB/T20438, also include the safety device of over speed/reversal protection/up and down step missing monitor protect function in SIL2 grade, and handrail/brake monitor function in SIL1 grade.

2 Overview and models defined

This document include the MONARCH integrated machine electric map and wiring map of T19.0 control system and the connection of the safety switches, have the explanation of the VVVF/autostart escalator and moving walk's installation, usage, maintenance, this document can be the user manual of the system installation, commission and maintenance.

The controller of the MONARCH integrated machine T19.0 control system is microcomputer board, dragging method is the VVVF.

1、Frequency integrated machine is MONARCH NICE2000。

2、The motor is single-winding motor, start winding delta connection; control the escalator speed by inverter frequency output。

3、Running mode:

— Simple running (energy saving is invalid): always check the rated speed running

— Quick stop cycle running: When there' s no passenger it stops, when there' s passenger, it can be one direction running.

— Speed cycle running: When there' s no passenger, the escalator running according to the 2/7 speed of the rated speed. When there' s passenger, keeps single direction running.

— Stop circulation speed: When there' s no passenger it the escalator running according to the 2/7 speed of the rated speed , after a period of time to stop, when there' s passenger, it can be one direction running.

3 Brief introduction of the control system

3.1 Out line

The control system includes drive station, control box, lighting, safety switches, control button, monitor device and connecting cables. This system according to the design and production national standard GB16899-2011

— This system use Frequency integrated machine program control; inverter expansion board is the center of the control system. The main controller parts and main drive control part are all in the controller box, its specific composition described please check the “control system components table”

— Power supply required is three-phase and five-wire, the voltage is 380V (+7%~ -7%) , the grounding resistance is not bigger than 4Ω.

3.2 Main functions

a) Normal running;

- b) Inspection running;
- c) intelligent frequency running;
- d) Intelligent frequency Auto start running(optional);
- e) Lack of phase, wrong phase protection;
- f) Motor overheating protection;

3.3 Assistant functions

- a) Safety brake;
- b) Central lubrication system;
- c) Escalators speed monitoring (over speeding);
- d) Non-manipulation reversed monitoring;
- e) Missing step monitoring;
- f) Fault display function;
- g) Handrail speed monitoring (under-speed);
- h) Brake condition monitoring;
- i) Braking distance monitoring;

Running direction indicator (intelligent frequency Auto start to must be equipped according to user requirements);

- j) Additional brake (lifting height greater than 6 meters or public transport type or according to user requirements)。

3.4 Safety circuit

- a) The connect point start from A06, go through Q7 to A7 and connect point start from A11, go through Q12 to A12 are basic circuit, can not jumper connect.
- b) The connect point start from A01, go through Q 1、 A02、 Q2、 Q3、 A03、 Q4、 A04、 Q5、 A05、 Q6 to A06 and connect point start from A07, go through Q8、 Q9、 A08、 A09、 A10、 Q10、 Q11 to A11, and connctet point start from A12, go through Q13、 A13、 Q14、 A14、 Q15、 A15、 Q16、 Q17、 Q18、 A16、 A17 to A18 are converntional safety circuit, when under inspeciton running condition, if need to solve the fault can do the jumper connection(Please shorting the wiring that have mark that is different witht the normal wire, do not use normal wire to do the jumper connection), but do not jumper connecting when the escalator under the normal running.
- c) When safety circuit and basic safety circuit are connected, then safety circuit contactor KC will connect, inverter expansion board safety circuit inspection input signal lights will on.

Note: When the jumper connection of the safety circuit is required during the inspection, the people must use the Debug Dedicated Short wiring, wiring with ordinary lines apparent inconsistency identity, this line must be removed after the repair work has been completed, to ensure the safety circuit.

3.5 Monitor

The safety function controller achieves the following monitoring function:

- a) Step missing monitoring;
- b) Escalator running speed monitoring(over speed);
- c) Non-manipulation reversed monitoring;

- d) Handrail speed monitoring (under-speed);
- e) Brake condition monitoring;
- f) Braking distance monitoring。

3.6 Safety protection device

1) Lack of phase, error phase protection of Power system

The power supply system lack of phase, error phase, and phase sequence protection relay, escalator can't run.

2) Motor overload protection

When the motor overload, increased main loop current, overload protection function, the escalator stop running.

3) Broken step chain protection

When the step chain has been over-stretched or broken, protection action, the escalator stop running.

4) Comb plate protection

When some foreign substance has been clipped in or between the combs, protection action, the escalator stops running.

5) Handrail inlet protection

When some foreign substance has been clipped in the handrail inlet ,protection action, the escalator stop running.

6) Step sagging protection

When any part of the step can't keep the space between the entrance and the comb plate, protection action, the escalator will stop running.

7) Skirt panel protection (optional)

When some foreign substance has been clipped in between the skirt panel and the step, protection action, the escalator stops running.

8) Speed monitoring device

When the speed reaches the rated speed of 120% or 80% of rated speed and non-operating reversed, the escalator stop running and cannot restart.

9) Emergency button

In the escalator entrance, under the head of automatic sidewalks, or in the middle part of the large height escalator is arranged at the emergency button.

10) Broken drive-chain protection (optional):

When the drive-chain has been over-stretched or broken, protection action, the escalator stop running.

11) Aux brake

With a motion feedback switch Aux brake, cut off safety circuit when Aux brake is action, only when the Aux brake reset to restart. Lifting height greater than 6 meter or public transport type must configure.

12) Step panel lighting

In the step top to bottom level transitional section, in order to avoid passenger

escalator step junction between the two steps of the risk of, the step at under a green lighting.

13) Step yellow border

To instruct passenger stepped on the yellow border range, yellow border outside prone to danger.

14) Handrail speed protection

The escalator normal running, when the speed is 15% lower than the rated speed and duration greater than 15 seconds, the escalator stop running, this function is invalid under inspection mode.

15) Escalator and moving walk step missing protection

The action protection when escalator and moving walk step missing, stop and cannot restart, enter fault locking system, this function is invalid under inspection mode.

16) Work brake

With a motion feedback switch work brake, escalator starts, brake system is not released, should prevent the start and enter fault locking system. When escalator running monitoring, in the brake instruction and band-type brake feedback logic is inconsistent, cut off the power supply safety circuit escalator make is stop, to enter the fault locking.

17) Static brush

Remove static in the running of the escalator.

18) Open the front panel protection

When open the front panel, limit switch or induction sensor action, the escalator stop running, this function is invalid under inspection mode.

19) Host jigger protection

When the inspection people need to jigger, limit switch action, the escalator cannot run.

20) Host overheating (optional)

When host overheating, the host internal temperature switch action, the escalator stop running

3.7 The advantage of the Auto start escalator

Under normal condition, the time of escalator loading people for running is only 50%, other 50% is empty loading time. After use the auto running mode, it avoid the 50% time of running time, make the escalator running time similar to the loading running time, the escalator actual running time is reduced, in that case it reduce the cost of the escalator running and reduce the spare parts friction, extend the using time of the escalator, compare as follow:

Without auto start

Set: the cost of the electricity is 0.7 KW/h, the motor 10kw, the escalator running for 10 hours/day.

Yearly cost of the electricity: $360\text{days} \times 10\text{hours} \times 10\text{KW} \times 0.7\text{yuan} = 25200\text{yuan}$

The maintenance cost: 3000yuan

Escalator service time: 10 years

With auto start

Then: Reduce the running cost: $25200/2=12600$ Yuan

Reduce the maintenance cost: $3000/2=1500$ Yuan

Extend the service time: $10 \times 2=20$ years

3.8 The controller of the escalator safety function

3.8.1 This safety function controller have over speed, reversal, step missing protection device, it according to the safety standard grade of GB16899-2011 and GB/T20438 and SIL2; It have handrail under speed protection and brake release monitor device, the system safety functions according to the safety standard GB16899-2011 and GB/T20438 and SIL1 as follow:

- Speeding 1.2 times protection: When monitor the escalator speed is 1.2 times faster than the rated speed, it will cut off the 1.2 times safety relay to cut off the work brake /safety circuit and the controller circuit power to make the escalator stops running.
- Speeding 1.4 times protection: When monitor the escalator speed is 1.4 times faster than the rated speed, it will cut off the 1.2 times and 1.4 times safety relay both to cut off the work brake/Aux brake/ safety circuit and the controller circuit power to make the escalator stops running.
- Reversal protection: Monitor escalator or moving walk ($\alpha > 6^\circ$) , when the running direction is same as provision direction, cut off the 1.2 times safety relay and 1.4 times safety relay to cut off the work brake/Aux brake/ safety circuit and the controller circuit power to make the escalator stops running.
- Step missing protection: Install at escalator driving and steering position, when inspect there's step missing, cut off the 1.2 times relay to cut off the work brake/Aux brake/ safety circuit and the controller circuit power to make the escalator stop before the gap(cased by the step missing) appear at the comb plate position.**(This function is invalid under inspection mode)**
- Handrail under speed protection: Monitor the handrail speed, when the speed is 15% lower than the rated speed, cut off the 1.2 times safety relay to cut off the work brake/Aux brake/ safety circuit and the controller circuit power to make the escalator stops running. **(This function is invalid under inspection mode)**
- Brake release monitor: When the escalator start running, the brake system do not release the escalator can not start running; Monitor the escalator running, when the order of the brake is different with the brake feed back, cut off the 1.2 times safety relay to cut off the work brake/Aux brake/ safety circuit and the controller circuit power to make the escalator stops running. **(This function is invalid under inspection mode)**
- Brake distance inspect:
If the brake distance exceed 1.2 times of the maximum distance(when the escalator and moving walk reated speed is 0.5m/s ,the Max brake distance is 1m, when the escalator and moving walk reated speed is 0.65m/s ,the Max brake distance is 1.3m), stop the escalator immidatly, should restart afte the fualt is lock is resetted. The inspection people should inspect and solve the system problem before fualt reset.

3.8.2 In order to satisfy the above requirement, should have the sensor system of monitor the escalator speed/running direction/step missing/brake distance/brake release; System of judge the above safety function, like calculate the escalator running speed/judge provision direction and

running direction is same or not. The systems that can make the escalator stop and keep in safety mode output. So to separate the systems into three subsystems: sensor/logic/actuators to meet the requirement of the safety standard. Check the detailed information in 《The user manual of the controller of escalator and moving walk safety standard》

4 Operation introduction

4.1 Escalator running mode

The driving of the escalator adjusts the speed by inverter; through adjust the power supply and voltage of the AC motor to change the revolution and change the escalator running speed, to reduce the running and maintenance cost; When there' s no passenger, after 60 seconds the escalator will running at the 2/7 of the rated speed. When there' s passenger, the escalator running at the rated speed.

4.2 Running analysis

a) Inspection running:

- (1)Unplug the Aux plug of the up or down control box, insert the inspection plug, relay KJX do not act, escalator will turn to inspection running, hand hold inspection box (The plug is multi-core aviation plug).
- (2)Check FU3—FU4 cabinet fuse, turn on the power switch.
- (3)The safety circuit is normal, inverter、expansion board with normal power supply, open the “STOP” switch on the inspection box, jog the public button “SQ” on inspection box and “UP/DOWN” button, the contactor will work according to the following order: The brake contactor KMB pull→running contactor YC pull→brake response switch KBZ1/KBZ2 act, escalator running according to the running mode on specification.
- (4)When the escalator under inspection mode, speed monitor/non-manipulated reversed/brake distance inspection are till working.

b) Normal running:

- (1)Insert the up/down inspection Aux plug, relay KJX pull.
- (2)Check FU3—FU4 cabinet fuse, connect power switch QF、K1、KF.
- (3)The safety circuit is normal, PLC with normal power supply, open the “STOP” switch on the inspection box, jog the public button “SQ” on inspection box and “UP/DOWN” button, the contactor will work according to the following order: the brake contactor KMB pull→running contactor YC pull→brake response switch KBZ1/KBZ2 act, escalator running according to the running mode on specification.

c) Intelligent VVVF running:

(1) Simple running mode:

This running mode is the standard running mode, no need intelligent sensor, when inverter parameter FB-16=0, escalator running according to the running mode on specification, until to stop button to stop.

(2) Quick stop cycle running:

When the inverter parameter FB-16=1, the escalator will start running. If in 60 seconds

there's no passenger go no passenger go into the sensor response area, the escalator will stop; when on the running direction there's passenger come, the escalator will running at the rated speed. When the escalator auto stop, if there's passenger come at the opposite running direction, the escalator will running in original economic running mode for 15 seconds and stop, (If in 15 seconds, there's passenger come in the right direction the escalator, running will at the rated speed), and inform the passenger he is in the wrong direction.

Note: the intelligent running time can set through inverter parameter.

(3) Speed cycle running:

When the inverter parameter FB-16=2, the escalator will start running. If there's no passenger go into the sensor response area, the escalator running according to the 2/7 speed of the rated speed. When there's passenger coming to the running direction, the escalator will run at the rated speed.

Note: the intelligent running time can set through inverter parameter.

(4) Stop circulation speed mode:

When inverter parameter FB-16=3, the escalator will start running. If in 60 seconds there's no passenger go no passenger go into the sensor response area, the escalator will running in economic running mode for 15 seconds and stop; when on the running direction, there's passenger come, the escalator will running at the rated speed. When the escalator stop, if there's passenger come at the opposite running direction, the escalator will running in original economic running mode for 15 seconds and stop (If in 15 seconds, there's passenger come in the right direction, the escalator), and inform the passenger he is in the wrong direction.

Note: the intelligent running time can set through inverter parameter.

4.3 Running director

At two sides of the escalator, there's running direction indicator, the passenger should use the escalator according the direction indicator; If it is green light, it is right direction for using the escalator.

4.4 Brake mode

(1)Emergency brake: During the escalator is running, when the stop button/safety circuit contacts/contacts act and step missing monitor/handrail speed monitor/handrail speed monitor/non-manipulation of anti-reversal/brake monitor act under normal running condition, the escalator will be emergency stop. At the same time, the brake act and will cut off the power supply of the motor.

(2)Order brake(Intelligent auto start stop escalator):

When the escalator under the single intelligent running mode, if there's no passenger come, then the escalator control system will cut off the power supply of the motor and stop the escalator running.

5 MONARCH integrated machine T19.0 Electrical commissioning instructions

5.1 (PLQ)、(PES) I/O definition

MONARCH integrated machine NICE2000 input/output (control/action) contacts setting conduction:

PLC Input	Signal Definitions	Condition
DC+	24V+ power supply	
DC-	24V- power supply	
X1	Normal/inspection running signal	Light on when running
X2	Up running signal	Light on when up running
X3	Down running signal	Light on when down running
X4	The contactor adhesions signal	Light dark when running
X5	The safety circuit inspect signal	Light on when safety circuit connect
X10	The up intelligent photoelectric sensor	Flicker
X11	The lower part of the intelligent photoelectric sensor	Flicker
PLC output	Signal Definitions	Condition
Y4	(YC) Delta contactor output	Running connect
Y6	(KMB) Brake contactor output	Running connect
Y7	(JDL) Bells output	Belling connect
Y8	(KUZ) Lubrication system output	Oiling connect
Y10	(SD) The up running direction signal	Up running connect
Y11	(XD) The down running direction signal	down running connect

PES SAFETY FUNCTION CONTROLLER INPORT/OUTPUT (CONTROL/ACTION)POINT SETTING CONDITION: PLEASE CHECK DETAILED ELECTRICAL MAP.

5.2 Escalator over speed\ reversal prevent safety function inspect

5.2.1 Escalator over speed protection function inspect

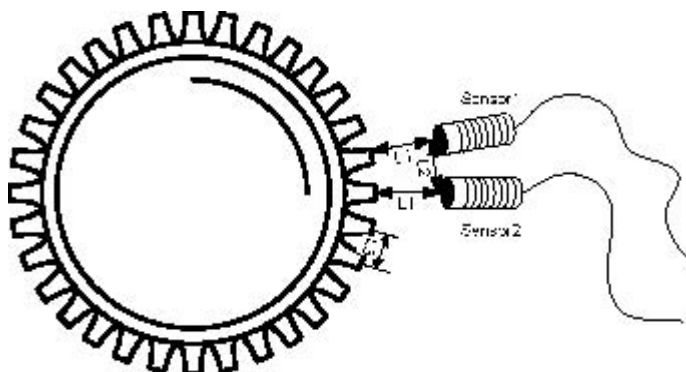
The principle of inspect this system is through inspect the speed of double sprocket to judge the escalator running speed is over speed or according to the safety function. Sensor 1、Sensor 2 (CS0、CS1) are inductive proximity switches. When the escalator is running, the teeth between the gears of the double sprockets go though the Sensor 1、Sensor 2 (CS0、CS1) and get pulse signal. The system calculates the real running speed of the escalator according to the pulse signal and compare with the rated speed to judge the escalator is over speed or not.

5.2.2 Escalator manipulation of non-anti-reversal protect inspect

The inspect principle:

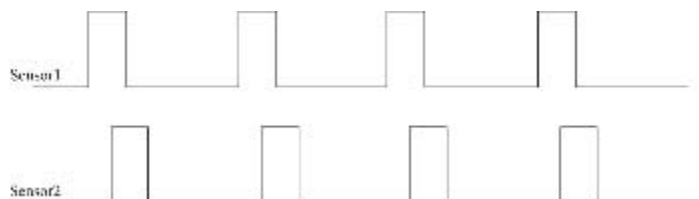
- 1, This system checking the running direction of the double sprockets and the given running direction to judge the protection function. Chain brake protect device can serial with escalator safety circuit when needed.
- 2, Sensor 1、Sensor 2 (CS0、CS1) are inductive proximity switches. Face the center position of the gear when install the Sensor 1 (CS0) , near the center position that face the nearby gear.
- 3, When the double sprocket running, Sensor 1、Sensor 2 (CS0、CS1) inspect the phase difference of the pulse. The escalator forward and reversion have opposite phase difference, from this compare with the given direction and do the reversal protection.

INSPECT ESCALATOR ACTUL RUNNONG SPEED/DIRECTION EFFECT DIAGRAM

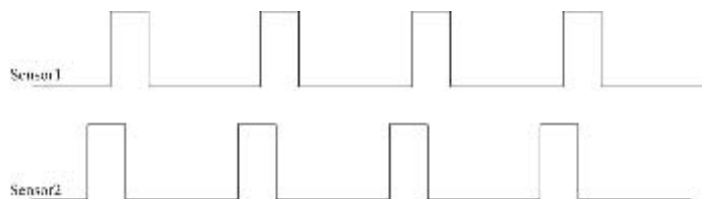


UP/DOWN INDUCTIVE PROXIMITY SWITCHES Sensor 1、Sensor 2 (CS0、CS1) EFFECT DIAGRAM

UP RUNNING DIRECTION



DOWN RUNNING DIRECTION



5.2.3 Brake distance function(Non safety function requirement)

Inspect principle: this system judge the brake switch feed back as the beginning of the inspection. When there's valid signal go through Sensor1 (CS0) and Sensor2 (CS1) to record the pulse number of the double sprockets N. When N bigger than the given brake distance, the brake distance function will be execution, the fault is fixed.

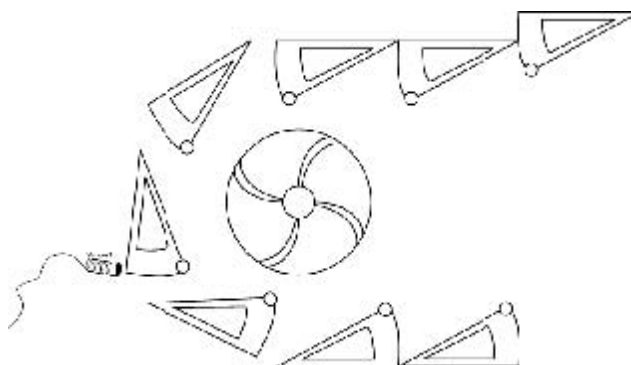
5.2.4 If the inductive proximity switches can not satisfy the action condition, so need the maintenance people to adjust. The voltage at site may have difference with the factory testing voltage, when the site power supply or voltage is lower will case escalator lack power when it is heavy load or full load running and make the escalator occur reversal and inverted slip, it is dangerous.

Note: Must make sure that the power supply should be stable, the fluctuation should between $(-7\% \sim +7\%)$.

5.3 Escalator step missing protect function inspect

5.3.1 Inspection principle:

- 1, This system inspect the step missing according to install the sensor at driving and corner of the escalator to finish this safety function.
- 2, Install a proximity sensor at driving and corner of up and down part of the escalator. When the escalator running, everything step pass will get a proximity pulse, if not receive the pulse after finish the distance of the step length, then the step missing function will be execution.



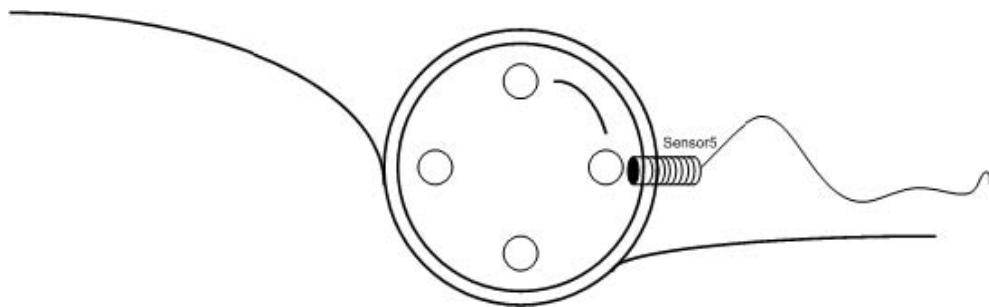
- 5.3.2 If the sensor cannot satisfy the above requirement, need the maintenance people to adjust the position of the proximity sensor.

5.4 Handrail under speed inspect

5.4.1 Inspection principle

1, This system execution the handrail under speed safety function through inspect the speed of handrail friction wheel and compare with the actual speed of the escalator running speed.

2, SCL and CSR are inductive proximity switches. Friction wheel have holes at each direction. When the escalator running, handrail fiction wheel running through friction. Sensor induction the holes on the friction wheel, form the pulse. Processing unit get the pulse signal and change to the handrail speed and compare with the speed of the step to complete the safety function.



Note: the friction wheel can have hole around, but must have holes uniform.

5.4.2 If the sensor cannot satisfy the above requirement, need maintenance people adjust the position of the sensor?

5.5 Method of inspect escalator over speed, reversal prevent, handrail under speed protect function

(1)Escalator over speed inspection: After the escalator is running under the rated speed, cover the inductive proximity sensors on the double sprockets, the system will cut off the safety circuit as the escalator over speed, stop the escalator, safety function controller cut off 1.2 or 1.4 times safety relay. This fault is power cut off remain, before solve the fault and reset the escalator can not running. Fault reset: push the reset button and save 3 seconds, then it will become normal.

(2)Non-manipulation of the anti-reversal inspection: Only the two speed inspection signal both participate the inspection and the phase difference is 90° , this function will be achieve. When the escalator is stop, change the wire 28 and 29 (means brake the definition phase difference), after it start running, the system will stop the escalator by itself, safety function controller brake 1.2 and 1.4 times safety relay, if there' s Aux brake, it will have power loss action. This fault is power off remain, before solve the problem and reset, escalator can not running. After change back the 28 and 29, the fault should be solve. Fault reset: push the reset button, keep 3 seconds, will become normal.

(3)Step missing inspection: Take one or more steps when the escalator is not running,

running the escalator till the escalator missing position, the escalator will stop, safety function controller will brake 1.2 times safety relay. This fault is power off remain, before solve the problem and reset, escalator can not running. After install the missing steps, the fault should be solve. Fault reset: push the reset button, keep 3 seconds, will become normal.

(4)The handrail under speed inspection: there's a proximity switch on the handrail, it monitor the running speed of the handrail. Under normal running, if the handrail speed is 15% lower than the step speed and keep 15 seconds, then the escalator will stop. When doing the inspection at site, the people can release the handrail or force some part slip to make it under speed, when the escalator reach the rated speed, it will inspect the handrail under speed and stop the escalator, safety function controller brake 1.2 times safety relay. This fault is power off remain, before solve the problem and reset, escalator can not running. Fault reset: push the reset button, keep 3 seconds, will become normal.

(5) The brake distance inspection: Release the brake when the escalator is stop, start the escalator, make the escalator running for some time and stop it, make it exceed the 1.2 times of max distance (when the escalator and moving walk reacted speed is 0.5m/s, the Max brake distance is 1m, when the escalator and moving walk reacted speed is 0.65m/s, the Max brake distance is 1.3m), safety function controller brake 1.2 times safety relay. This fault is power off remain, before solve the problem and reset, escalator can not running. Fault reset: push the reset button, keep 3 seconds, will become normal.

Note: When inspect at site, make sure the safety; after inspection should restore all the signal wire and sensor to normal condition.

5.6 Fault indicator board

(1)The voltage of the fault indicator is AC8V、AC110V.

(2)When the fault indicator indicate “□”, it means the safety circuit are all connected.

(3)When the safety circuit action, the fault indicator indicate the related the fault code, fault code please check the code table.

5.7 Send out, transport and site work

After the escalator send out the factory, because of the transport, lifting, working condition at site, before installation and use, the escalator should adjust the distance between the skirting panel and switch, front panel and the switch, comb switch, step missing sensor, escalator speed sensor and handrail speed sensor and other electrical parts.

5.8 Other introduction

5.9.1 Brake release protection: there's brake release switch on the work brake, when the brake lose power and escalator stop, brake release switch is OFF, when the brake is with power supply, the escalator is running, the brake release switch is ON. If when the escalator start and the brake release switch is OFF, then the escalator can not run; when the escalator running and inspect the brake release signal is OFF, escalator stop, protection condition lock itself, only through fault reset the brake

release protect can cancel.

5.9.2 Contactor adhesion protection: After the system stop the escalator (normal stop and abnormal stop), all contactors are in the position of release, expansion board inspection signal should be ON, or the system will go into protection mode, escalator can not start.

5.9.3 If the data setting of the safety function controller is not correct, the escalator can not running normally, please check the data setting is same with escalator/moving walk.

5.9.4 If specification of the escalator/moving walk do not included in the safety function controller setting range, then the safety device maybe not suitable for the escalator /moving walk.

6 Electrical maintain introduction

6.1 Escalator daily maintenance

In order to insure the normal running of the escalator and extend the service life, we should do regular maintenance of the escalators:

—— Every day should do once maintenance:

- 1) Every day should do the cleaning of the step, comb.
- 2) Clean the handrail/glass and other place that can easily see.
- 3) Check the out look of the escalator, if there any damage, should take care.
- 4) Every time start the escalator, should listen to the voice of the escalator, if there' s abnormal sound.
- 5) There must be the person responsible for the escalator, set up the running record of the escalator, in order to check and handle the management of the escalator.

—— The escalator must check and adjust once after it running for three month.

- 1) Check and adjust the safety switches.
- 2) Clean the control screen and related connectors.
- 3) Check the ground terminal and the static brushes.

——Every half month, should clean the up/down machine room.

——Every month, should check the oiling device have oil or not and fulfil the oil.

If there is oiling device, the inverter expansion board will accumulated working time and when the time is reach 24 hours, will make the oiling device oiling for 3 minutes, the user and set the oiling time and frequency of the escalator according to condition; If under some condition, need to oiling manually(the oiling device is electromagnetic pump), When the oiling device is gear pumps, the oiling can do according to the maintenance requirement by push button.

Repair is the overhaul according to the plan, after repair the equipment should meet the technical requirement before the new equipment sending out the factory. If the equipment using for long time and some aspects can not reach the requirement but can running safely, we can release the requirement, but the

inspection work should be carefully.

The safety rules and precaution during inspection work:

- 1) Prepare the inspection space; clean the working channel for the inspection people.
- 2) The should have safety fencing and warning signs like “The escalator is under maintenance” ;
- 3) Before inspection, should cut off the electrical master switch and put on the warning signs “Under inspection” ;
- 4) People can run the escalator only by disc wheel.
- 5) When the inspection work is finish, should check carefully there can not leave any tools or other things in the escalator, especially between the relative motions spare parts.

6.2 Simple maintenance method

- (1) If the escalator cannot start running, please check the power supply and the voltage is normal or not firstly. If normal, check is there any fault code or not. If there' s not fault code, please check as follow.
- (2) The expansion board output point and the expansion board is normal or not.
- (3) Check the contactor adhesions signal expansion board input point is ON or not, check the circuit is normal or not.
- (4) Check the inverter is normal or not.
- (5) If the above circuit is all normal, check the up (down) running start signal expansion board input point is ON or not. Otherwise check the start-up circuit.
- (6) If (1)–(5) are all normal, escalator still can not start, then cut off the power supply and restart; If after restart, still not running, should do the checking as follow:
 - ① Escalators over speed/ non-manipulation of the anti-reversal/step missing/brake release/brake distance self lock protection release, and then restart the escalator. If the escalator can start, then the reason of not running is cased by escalators over speed/ non-manipulation of the anti-reversal/step missing/brake release/brake distance self lock protection release; If after the above operation, the escalator still can not start, please check the wiring according to the wiring map.
 - ②If the escalator does not have the intelligent start running function, please check the escalator as follow: First make sure there' s nothing cover the sensor and have no signal input.
- (6) Any problem related to the inspection of the escalator over speed/step missing/brake release/handrail under speed/brake distance inspect please solve refer to 《The user manual of the safety function control device》.

Dangerous: The people do not professional training before are strictly prohibit to do the maintenance of the escalator; Unskilled or people who are not electrician are allowed to do the work to make sure people' s safety.

7 Control system element form

Code	Name	Model & Specification	Number	Position
KMB	Brake contactor	AC110V Fuji/Schneider/Siemens	1	Up control box
YC	Delta running contactor	AC110V Fuji/Schneider/Siemens	1	Up control box
KJX	Inspection relay	AC110V Fuji/Schneider/Siemens	1	Up control box
KPH	Phase relay	SW11	1	Up control box
FRT/FRT1	Motor thermal relay	3UA59	1	Up control box
FU3-FU4	Fuse	2A	3	Up control box
XS1-U	Three flat socket	AC220V	1	Up control box
XS1-D	Three flat socket	AC220V	1	Down control box
XS2-U	Two flat socket	AC220V	1	Up control box
XS2-D	Two flat socket	AC220V		Down control box
XS	Fault indicator	XS-A 或 XS-B	1	Up control box
QF	Leakage break protection switch	DZX2-C63	1	Up control box
KF	Circuit brake protection switch	DZX2-C6	1	Up control box
K1	Main power switch	JFD11	1	Up control box
K2	Illumination switch	DZX2-C6	1	Up control box
KC	Safety circuit contactor	AC110V Fuji/Schneider/Siemens	1	Up control box
B1	Working transformer	BK-200AC380V/AV8V、110V	1	Down control box
B2	36V Illumination power	KBY-01-36V	1	Down control box
M1/M2	AC motor	AC380 Three-phase	1	Up machine room
M3/M4	Brake motor	AC380 Three-phase	1	Up machine room
ZDQ/ZDQ1	Brake motor	AC220V Single-phase	1	Up machine room
JZD/JZD1	Aux brake	AC110V/AC220V/AC380V	1	Up machine room
XPI-U	Inspection Aux plug	WS16 Plug	1	Up inspect

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				outlet
XPI-D	Inspection Aux plug	WS16 plug	1	Down inspect outlet
JUZ	Automatic oiling device		1	In truss
SA	Oiling manual switch		1	On oiling device
XIB	Inspection operating handle	YT01-C/E	1	On inspect handle device
XIP	Inspection plug	WS16 Plug	1	On inspect handle device
SXI-U	Inspection outlet	WS16 Outlet	1	In up controller box
SXI-D	Inspection outlet	WS16 Outlet	1	In down controller box
SBSTP-IU	Up control box emergency button	LAY series	1	In up controller box
SBSTP-ID	Down control box emergency button	LAY series	1	In down controller box
SBSTP-U	Up emergency stop button	LAY series	1	At up part of the escalator
SBSTP-D	Down emergency stop button	LAY series	1	At down part of the escalator
SRST-U	Up key switch	LAY series	1	At up part of the escalator
SRST-D	Down key switch	LAY series	1	At down part of the escalator
SRST1-U	Up running button	LAY series	1	On inspect handle device
SRST1-D	Down running button	LAY series	1	On inspect handle device
SQ	Public button	LAY series	1	On inspect handle device
SBST	Emergency stop button	LAY series	1	On inspect handle device
RC1—RC6	RC absorption module	HD1.211	6	On up controller box

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KR	Heating contactor	LC1-D32F7C AC220V	1	On up controller box
JR1-JRn	Heating resistance	1000W	n	In truss
EF1...EFn	Skirting panel lighting	DC24V	n	On skirting panel
FUR, FUL	Handrail lighting	DC24V	2	Handrail support
DER、DEL	Down left/right comb lighting	DC24V	2	On skirting panel
UER、UEL	UP left/right comb lighting	DC24V	2	On skirting panel
HDL	Down running indicator	DC24V	1	On down truss
HUL	Up running indicator	DC24V	1	On up truss
UL	Up step lighting	AC220 LED	1	In up truss
DL	Down step lighting	AC220 LED	1	In down truss
SWK	Water level switch	C61F-GP	1	In down controller box
FS	Control box fan	AC220V	1	In up controller box
WKQ	Temperature controller	TC4S-24R	1	In up controller box
SBSTP-FD	Down Aux emergency stop button	LAY series	1	On down truss
SBSTP-FU	Up Aux emergency stop button	LAY series	1	On up truss
FSD/FSD1	Motor disc wheel switch	TS236	1	On motor
RSD/RSD1	Motor thermal switch		1	In motor
WMD	Down front board open switch	TR236	1	In down escalator truss
WMU	Up front board open switch	TR236	1	In up escalator truss
KDZ	Aux brake inspect switch	TS236	1	In up escalator truss
KZD	Aux brake relay	HHC68B-2Z HH54P AC220V	1	In up controller box
KDL	Bell	AC220V	1	In up

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				controller box
TPB	Switch power	/ NES-35-24/ NES-50-24	1	In up controller box
TPB1	Switch power	NES-35-24	1	In up controller box
TPB2	Switch power	LRS-100-24/LRS-200-24	1	In up controller box
SAHR-UL	Upper left entrance safety switch	TR236	1	In the upper right entrance
SAHR-UR	Upper right entrance safety switch	TR236	1	In the up left entrance
SAHR-DL	Lower left entrance safety switch	TR236	1	In the lower right entrance
SAHR-DR	Lower right entrance safety switch	TR236	1	In the down left entrance
SASP-UL	Upper left skirting panel safety switch	TR236	1	Near the inside of up right skirting panel
SASP-UR	Upper right skirting panel safety switch	TR236	1	Near the inside of up left skirting panel
SASP-DL	lower left skirting panel safety switch	TR236	1	Near the inside of down right skirting panel
SASP-DR	Lower right skirting panel safety switch	TR236	1	Near the inside of down left skirting panel
SADR/SADR1	Drive chain safety switch	ZR236 or UKS	1	Near the drive chain
SACR-DL	Left step chain safety switch	ZR236 or UKS	1	On right chain tension device
SACR-DR	Right step chain safety switch	ZR236 or UKS	1	On left chain tension device
SACP-UL	Upper left comb abnormal safety switch	ZR236 or UKS	1	On left of the up comb plate
SACP-UR	Upper down comb abnormal safety switch	ZR236 or UKS	1	On right of the up comb plate
SACP-DL	lower left comb abnormal	ZR236 or UKS	1	On left of the

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	safety switch			down comb plate
SACP-DR	Lower right comb abnormal safety switch	ZR236 or UKS	1	On right of the down comb plate
SADD-U	Upper step sag safety switch	ZR236 or UKS	1	On up step bend truss
SADD-D	Lower step sag safety switch	ZR236 or UKS	1	On down step bend truss
KBZ1/KBZ2	Brake release inspect switch	TS236	2	On motor
PES	Safety function controller	MCTC-PES-E1	1	Up controller box
SRES	Safety function fault reset button		1	Up controller box
CSL	Right handrail speed inspect sensor	Omron/Schneider/Huitong series	1	On tension device of the right handrail
CSR	Left handrail speed inspect sensor	Omron/Schneider/Huitong series	1	On tension device of the left handrail
CS0	Speed monitor sensor1	Omron/Schneider/Huitong series	1	In escalator truss
CS1	Speed monitor sensor2	Omron/Schneider/Huitong series	1	In escalator truss
CSD	Lower step missing sensor	Omron/Schneider/Huitong series	1	In escalator truss
CSU	Upper step missing sensor	Omron/Schneider/Huitong series	1	In escalator truss
CS2	Lower auto start sensor	CZ-111B	1	In escalator truss
CS3	Upper auto start sensor	CZ-111B	1	In escalator truss
INV	inverter	MONARCH NICE2000		Inverter control box
PLQ	Inverter expansion board	MONARCH MCTC-KZ-C		Inverter control box
JDL	Bell relay	HHC68B-2Z HH52P AC110V		Up controller box
KUZ	Oil relay	HHC68B-2Z HH52P AC110V		Up controller box

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BR	Brake resistance	1700W-15000W		Inverter control box
UK1	Down front board open switch	TS236	1	In escalator truss
UK2	Down front board open switch	TS236	1	In escalator truss
DK1	Down front board open switch	TS236	1	In escalator truss
DK2	Down front board open switch	TS236	1	In escalator truss
BAS	Network monitoring	C2000 M281/FL-IO-0606	1	Up controller box

8 Fault code form

Safety circuit fault code form

a. The fault code form of fault indicator on the skirting panel:

Code	Name of the safety switch	Code number
☐	Safety circuit normal	
1	Safety circuit ground protection	
2	Drive chain brake switch abnormal	SADR
3	Water level switch/left & right step chain switch abnormal	SWK、SACR-DL、SACR-DR
4	Down left & right comb switch abnormal	SACP-DL、SACP-DR
5	Down left & right handrail entrance switch abnormal	SAHR-DL、SAHR-DR
6	Down left & right skirting panel switch abnormal	SASP-DL、SASP-DR
7	Down stop button/Emergency stop on down control box	SBSTP-D、SBSTP-ID
8	Inspection outlet of down control box/down Aux brake/down front board switch abnormal	SXI-D、SBSTP-FD、WMD
9	Down step sag switch abnormal	SADD-D

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10	Up step sag switch abnormal	SADD-U
11	Up front board/Aux brake/control box inspect outlet abnormal	WMU、SBSTP-FU、SXI-U
12	Stop of the up control box/ up stop button	SBSTP-IU、SBSTP-U
13	Up left & right skirting panel switch abnormal	SASP-UL 、SASP-UR
14	Up left & right handrail entrance switch abnormal	SAHR-UL、SAHR-UR
15	Up left & right comb switch abnormal	SACP-UL 、SACP-UR
16	Motor disc wheel switch/motor thermal switch	FSD、RSD
17	Motor thermal protection/phase sequence protection/Firefighting linkage abnormal	FRT 、 KPH、 XF
18	Safety function controller fault	PES

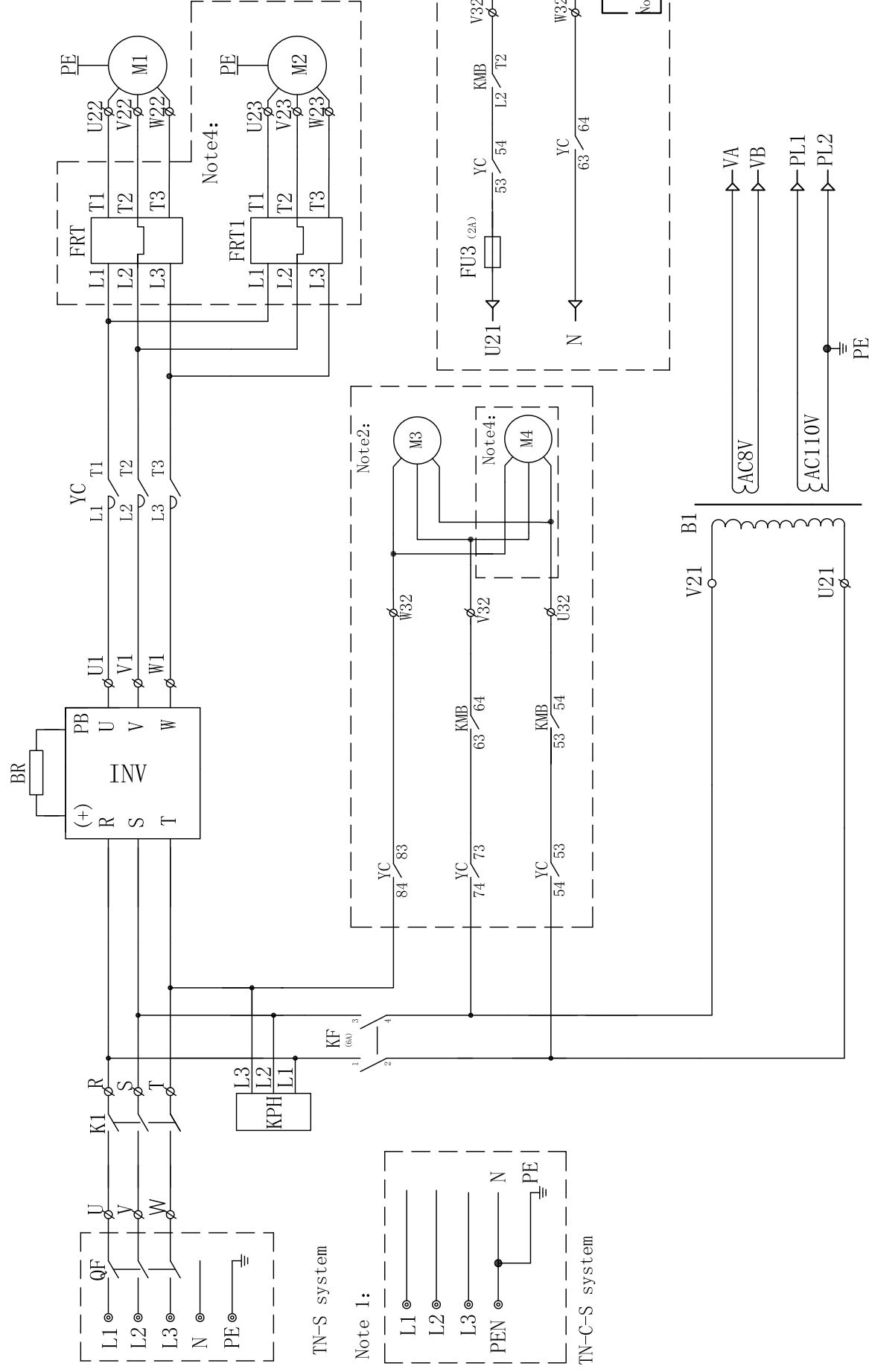
b: The fault code form of fault indicator on the control box:

Code	Name of the safety switch	Code number
☐	Safety circuit normal	
E29	Safety circuit ground protection/Drive chain brake switch abnormal	SADR
E28	Water level switch/left & right step chain switch abnormal	SWK、SACR-DL 、SACR-DR
E27	Down left & right comb switch abnormal	SACP-DL 、SACP-DR
E26	Down left & right handrail entrance switch abnormal	SAHR-DL、SAHR-DR
E25	Down left & right skirting panel switch abnormal	SASP-DL SASP-DR
E24	Down stop button/Emergency stop on down control box	SBSTP-D、SBSTP-ID
E23	Inspection outlet of down control box/down Aux brake/down front board switch abnormal	SXI-D 、SBSTP-FD、WMD
E22	Down step sag switch abnormal	SADD-D
E21	Up step sag switch abnormal	SADD-U
E19	Up front board/Aux brake/control box inspect outlet abnormal	WMU、SBSTP-FU、SXI-U
E18	Stop of the up control box/ up stop button	SBSTP-IU、SBSTP-U
E17	Up left & right skirting panel switch abnormal	SASP-UL 、SASP-UR
E16	Up left & right handrail entrance switch abnormal	SAHR-UL、SAHR-UR
E14	Up left & right comb switch abnormal	SACP-UL 、SACP-UR
E13	Motor disc wheel switch/motor thermal switch	FSD、RSD

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E12	Motor thermal protection/phase sequence protection/Firefighting linkage abnormal	FRT 、 KPH、 XF
E11	Safety function controller fault	PES

9 Electrical map and wiring diagram:



Notel :The best grounding for power supply is TN-S system named three phase five wire system or TN-C-S system undermost requirement make sure that the neutral wire is always separated from grounding wire as soon as power

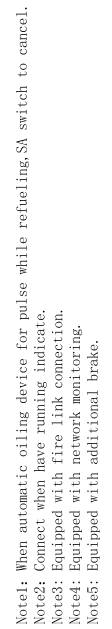
wife
entering machine room.

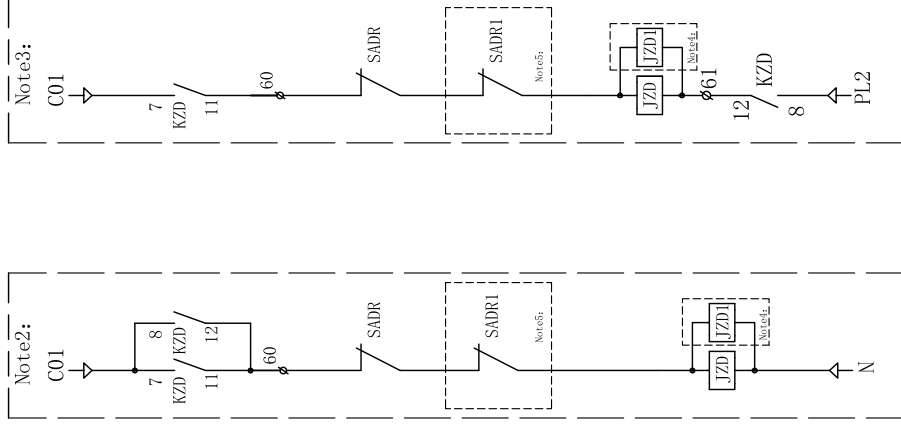
Note2 : Use it when contracting brake monitor coil for three phase AC380V.

Note3 : Use it when contracting brake monitor coil for one phase AC220V.

Explain: This drawing is according to the standard of MONARCH NICE2000 inverter & Schneider contactor.

Designed		Checked		Title	POWER CIRCUIT
				Drawing NO.	VFT-T19_0
Drawn		Date		Page NO.	1/10





Note 1: Connect when have AUX brake.

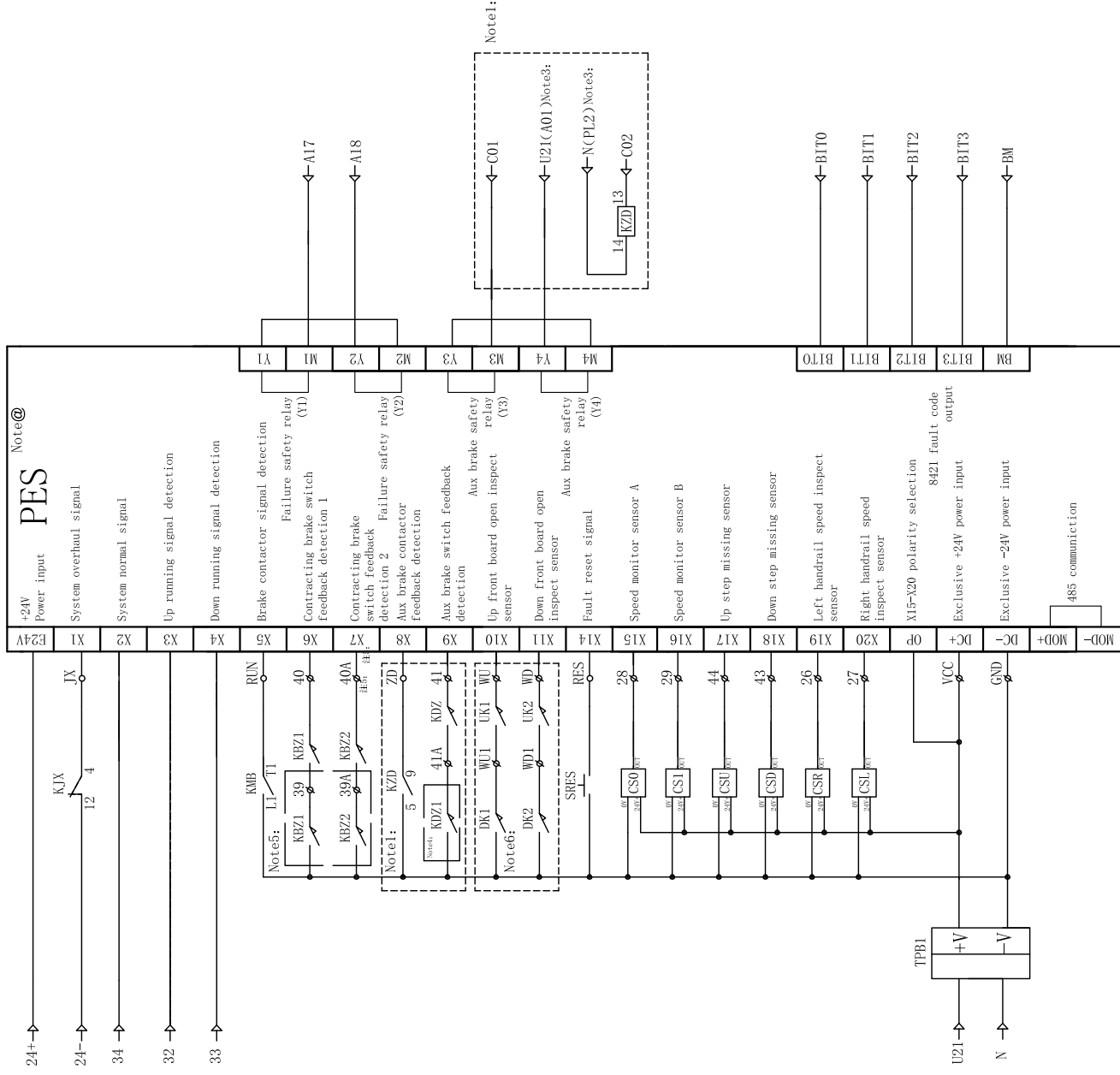
Note 2: Use it when Aux brake coil for AC220V.

Note 3: Use it when Aux brake coil for AC110V.

Note 4: Use it when it's double AUX brake.

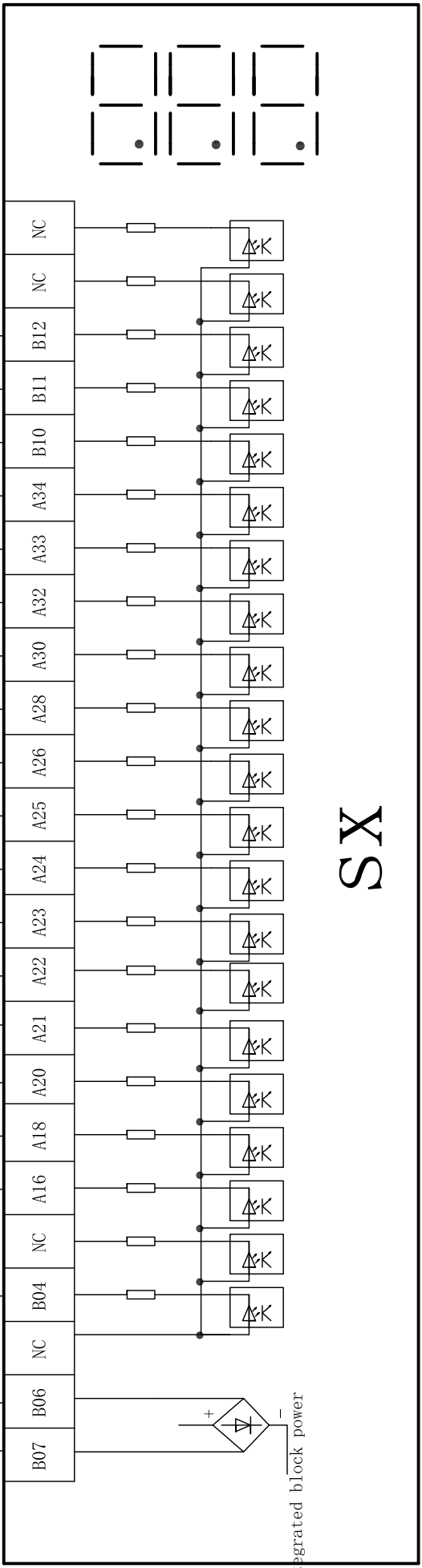
Note 5: Use it when it's double drive.

Note 6: If use safety switch the WU or WD signal corresponding connection to GND.

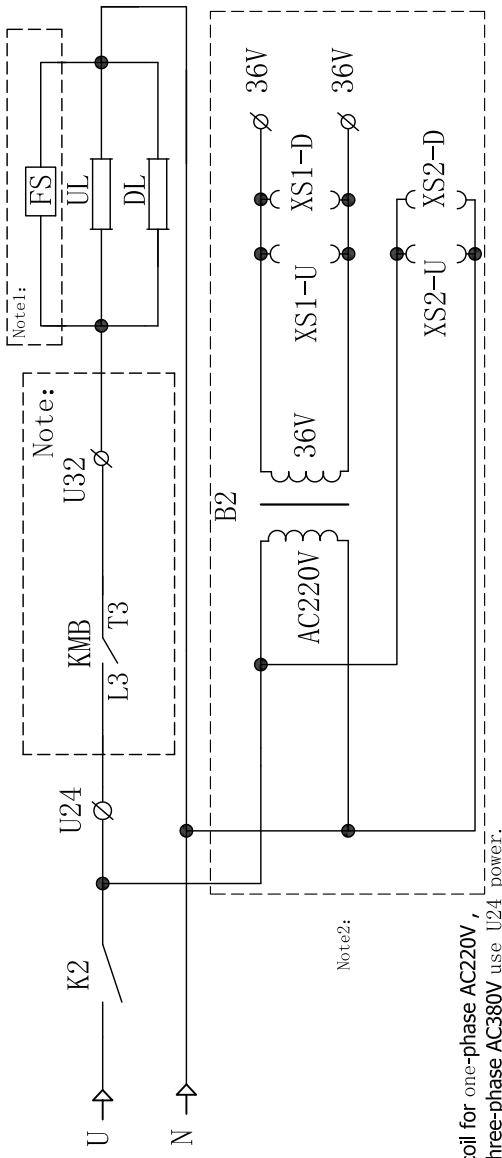


Note @: Use this diagram connection when the safety function controller is Monarch MCTC-PES-E1.

Designed		Checked	Title	Drawing No.	VFT-T19.0
Drawn		Date		Page No.	3/10



XS



Note: Use it when contracting brake motor coil for one-phase AC220V, when contracting brake motor coil for three-phase AC380V use U2

Note 1: When the escalator for outdoor use it

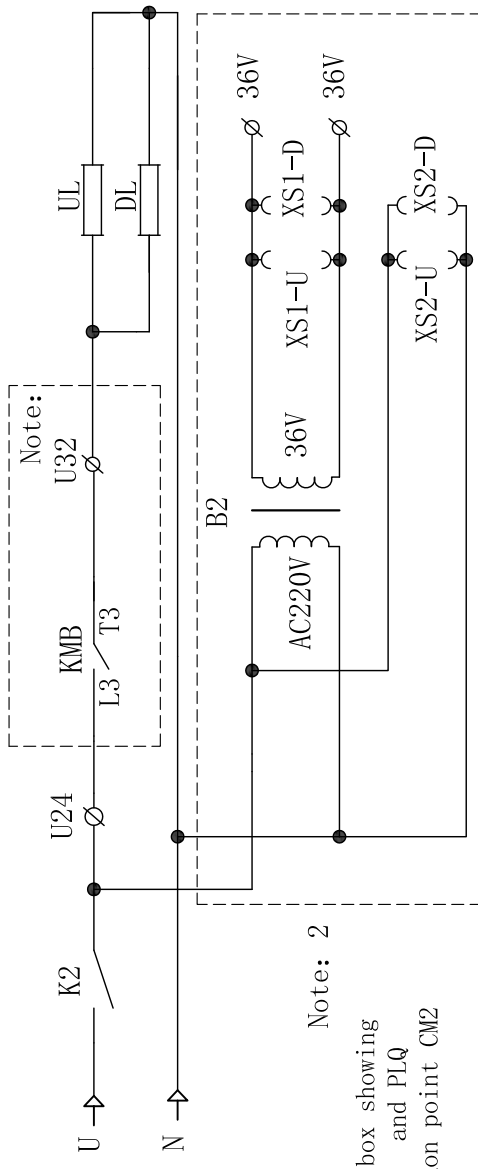
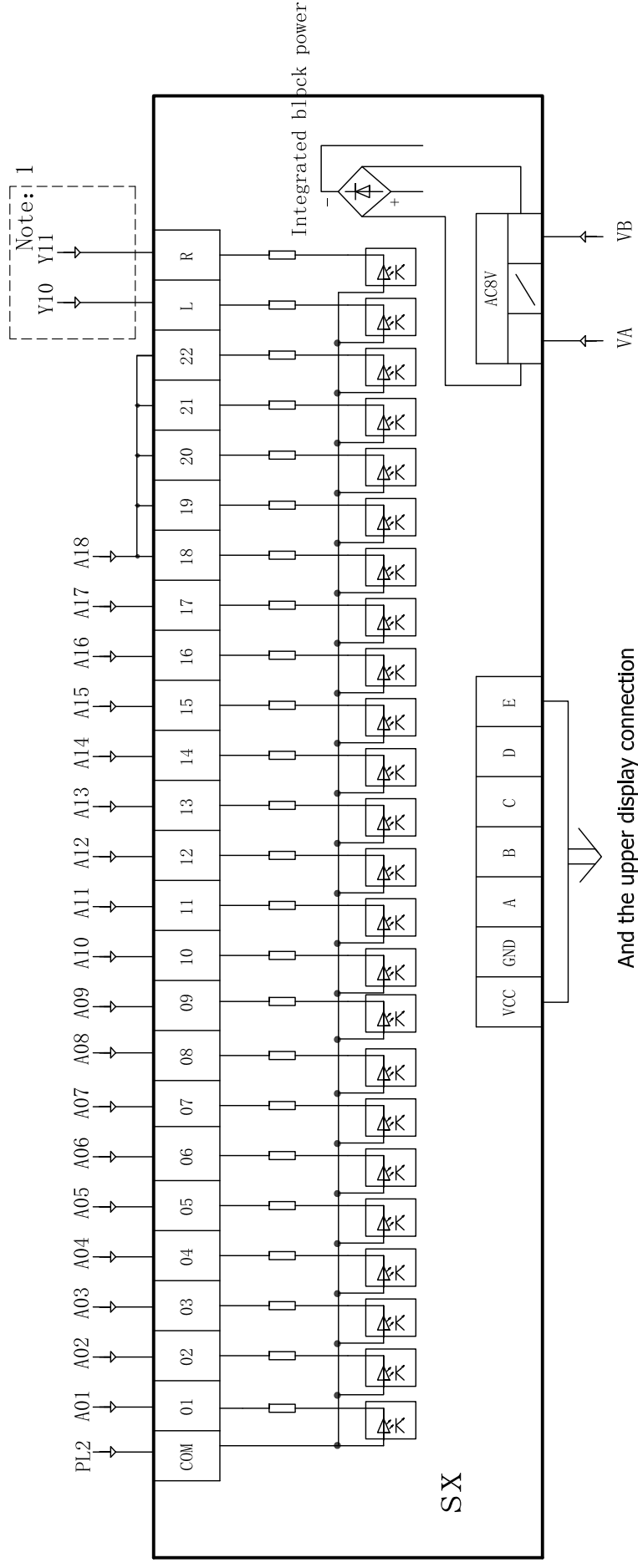
Note 2: 36V lighting Option

@: Use this diagram connection when fault display installed in control.

Explain:

When the fault display board internal photoelectric coupling input breakdown conduction, fault indication is not normal, but safety circuit function normal.

						SAFETY CIRCUIT (CONTROL BOX DISPLAY)
Designed	Checked		Title	Drawing No.	VFT-T19.0	
Drawn	Date		Page No.	5/10		



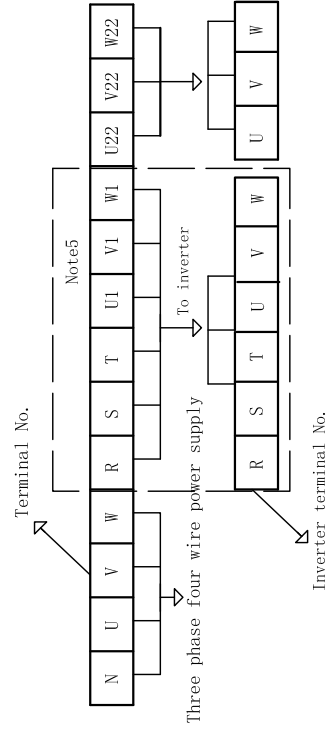
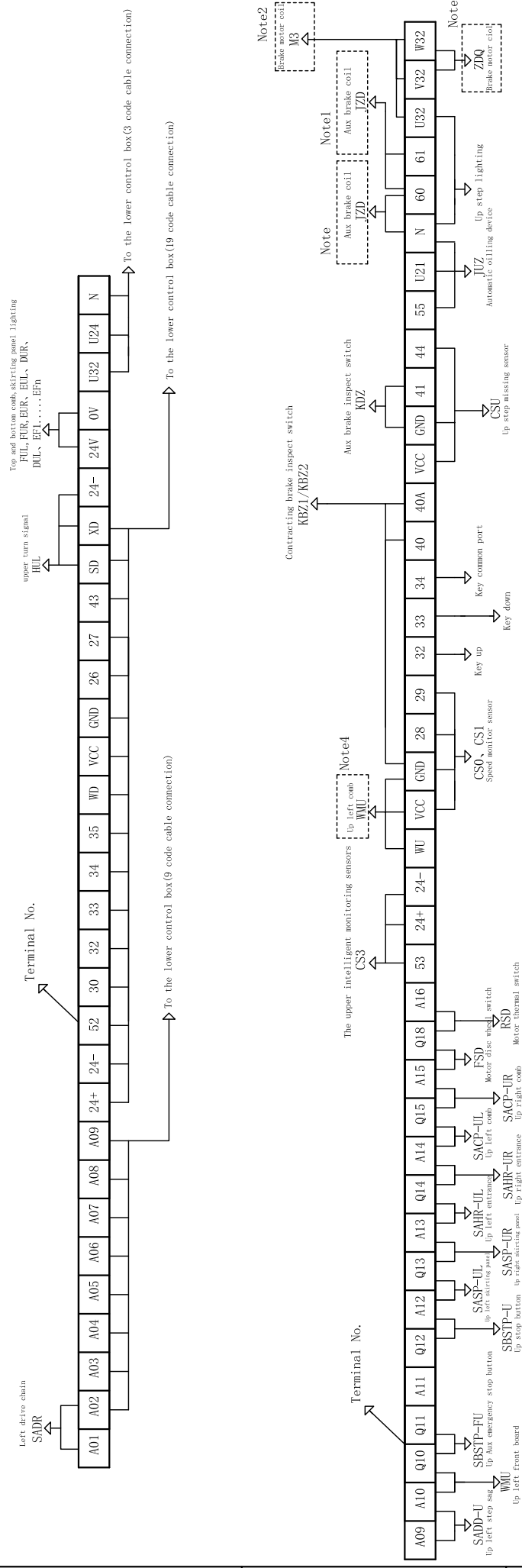
Note1: There is a skirt key box showing the direction of operation, and PLQ extension board output common point CM2 to A01.

Note: Use it when contracting brake motor coil for one-phase AC220V ,
when contracting brake motor coil for three-phase AC380V use U24 power.

Note 2: 36V lighting option

@: Use this diagram connection when fault display installed in skirting panel.

					SAFETY CIRCUIT(SKIRTING PANEL DISPLAY)
Designed		Checked		Title	
				Drawing No.	VFT-T19.0
Drawn		Date		Page No.	5/10



Note: Use it when Aux brake coil for AC220V.

Notel:Use it when Aux brake coil for AC110V.

Note2:Use it Contracting brake motor coil is AC380V, and according to the AC380V circuit wiring.

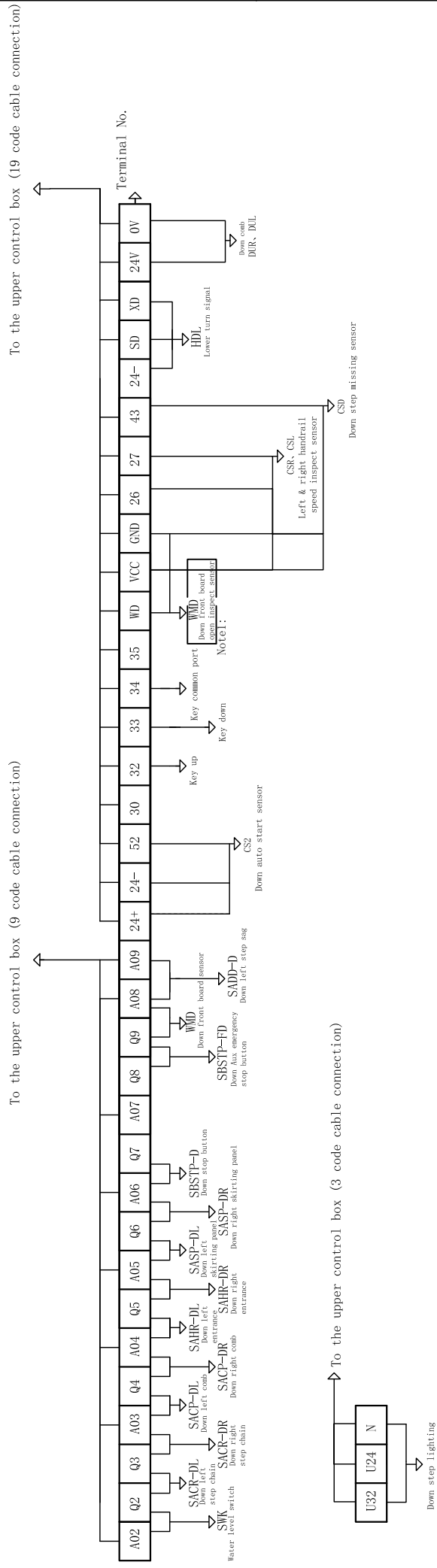
Note3:Use it Contracting brake motor coil is AC220V, and according to the AC220V circuit wiring.

Note4:If the front board open switch is safety switch, and series to safety circuit, the contact to cancel.

Note5: When frequency conversion split cabinet.

[illegible]

Technical specification: The drawing's guideline is the copper core cable.



A02	A03	A04	A05	A06	A07	A08	A09	Terminal No.
1	2	3	4	5	6	7	8	9 code cable

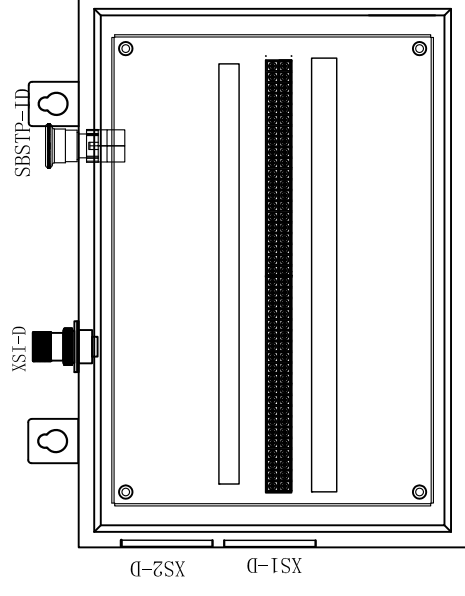
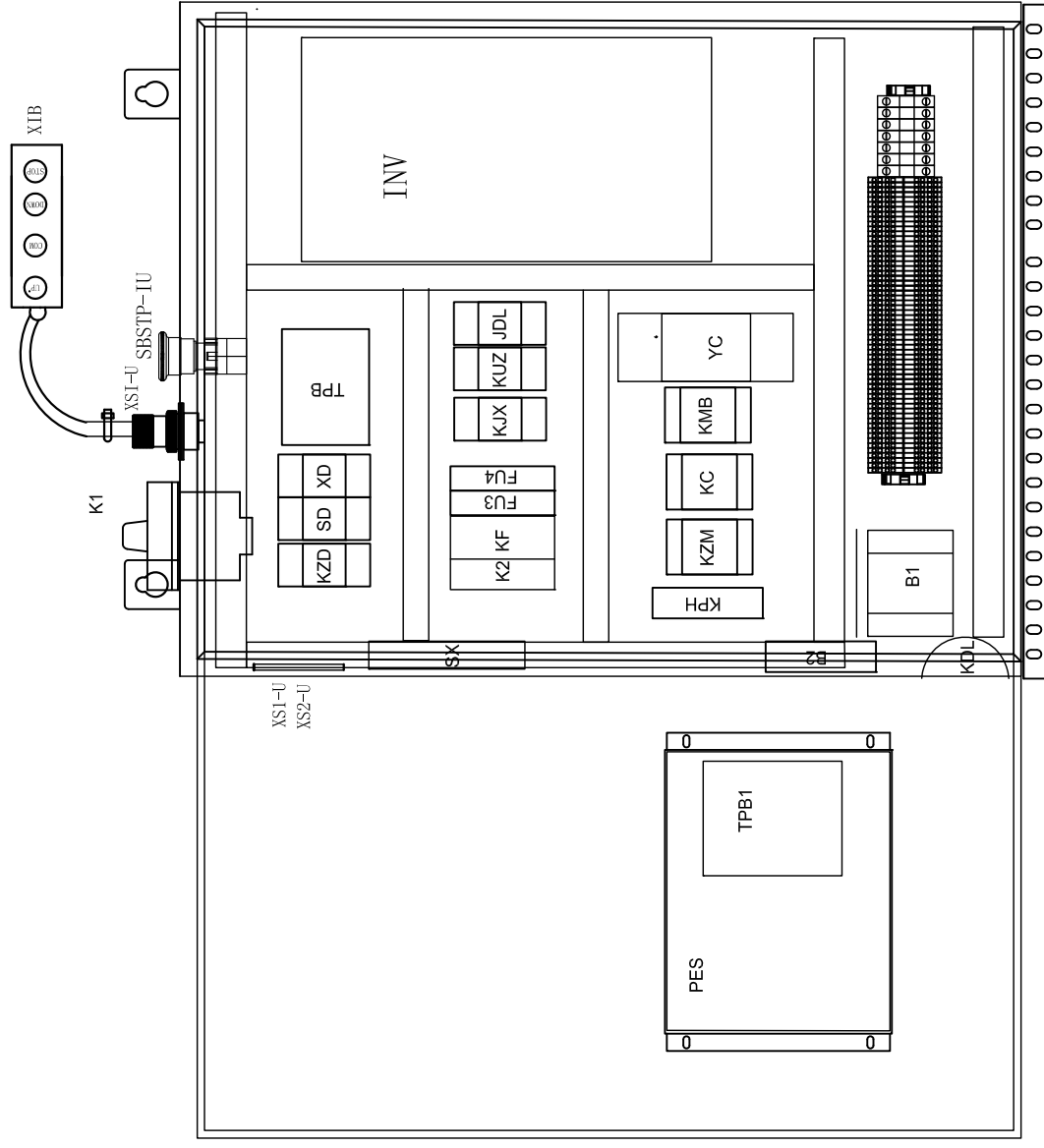
24+	24-	52	30	32	33	34	35	WD	VCC	GND	26	27	43	SD	XD	24V	0V	→Terminal No.
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19 →19 code cable number

U32	U24	N	Terminal No.
1	2	3	3 code cable number

Technical specification:
The drawing's guideline is the copper core cable.

Note: If the front board open switch is safety switch, and series to safety circuit, the contact to cancel.

Designed		Checked		Title	THE LOWER CONTROL BOX ELECTRICAL WIRING DIAGRAM
				Drawing No.	VFT-T19. 0
Drawn		Date		Page No.	9/10



Technical requirement: This map only for the reference of the control panel assemble, the position of electrical parts on the map may not accord with the real ones. when do the installation, people should adjust the position of the electrical parts and make sure the electrical parts are fixed and action reliable.

[illegible]

Preface

Thank you for purchasing the MCTC-PES-E1 system.

MCTC-PES-E1, independently developed and produced by Suzhou Monarch Control Technology Co., Ltd., is a programmable electronic system in safety related applications for escalators and moving walkways (PESSRAE).

The MCTC-PES-E1 adopts dual-CPU control and provides multiple security monitoring protections for escalators and moving walkways. The system satisfies the GB/16899-2011 Safety Rules for the Construction and Installation of Escalators and Moving Walks.

This manual describes the correct use of the MCTC-PES-E1 system. System installation, commissioning and maintenance must be performed only by qualified personnel with related experience.

It introduces the construction, features, security prompts, design & installation, operation & maintenance, and faults & solutions of the MCTC-PES-E1 system. Read and understand the manual before use, and keep it properly for future reference.

Notes
• The drawings in the manual are sometimes shown without covers or protective guards. Remember to install the covers or protective guards as specified first, and then perform operations in accordance with the instructions. • The drawings in the manual are shown for description only and may not match the product you purchased. • The instructions are subject to change, without notice, due to product upgrade, specification modification as well as efforts to increase the accuracy and convenience of the manual. • Contact our agents or customer service center if you have some questions during the use.

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1

Safety Information and Precautions

Chapter 1 Safety Information and Precautions

1.1 Safety Precautions

Use Stage	Safety Grade	Precautions
During installation	 DANGER	- Do not install the equipment if you find water seepage, component missing or damage upon unpacking. - Do not install the equipment if the packing list does not conform to the product you received. - Install the equipment on incombustible objects such as metal, and keep it away from combustible materials. Failure to comply may result in a fire. - Do not loosen the fixed screws of the components.
	 WARNING	- Handle the equipment with care during transportation to prevent damage to the equipment. - Do not use the equipment with damaged or missing components. Failure to comply will result in personal injury. - Do not touch the components with your hands. Failure to comply will result in static electricity damage. - Install the equipment in places free of vibration and direct sunlight.
At wiring	 DANGER	- Wiring must be performed only by qualified personnel under instructions described in this manual. Failure to comply may result in unexpected accidents. - Ensure that the power supply is cut off before wiring. Failure to comply may result in electric shock. - Before wiring, cut off the power supply of the escalator or moving walkway using reliable insulation device. Do not guarantee safety of engineers by depending on the stop, startup or input signal of the equipment because these signals cannot insulate hazardous voltage.
	 WARNING	- Pay attention to the marks of the wiring terminals and ensure correct wiring. Failure to comply will result in damage to the controller. - Use a shielded twisted pair (STP) cable with twist pitch of 20–30 mm for communication, and ensure that the shield is reliably grounded.

Use Stage	Safety Grade	Precautions
During operation	 DANGER	- Perform wiring of all peripheral devices under the instructions of the manual and ensure correct wiring according to the circuit diagram in the manual. Failure to comply will result in accidents. - Do not touch the system or surrounding circuits. Failure to comply will result in electric shock. - Do not touch any I/O terminal of the system. Failure to comply may result in electric shock. - Non-qualified personnel are prohibited to perform signal detection during operation. Failure to comply may result in personal injury or damage to the equipment.
	 DANGER	The equipment has high voltage input. The hazardous voltage may result in severe electric shock or even death. Please keep cautious when you use or get close to the system that has been connected to the power supply.
	 WARNING	- Ensure correct wiring of input and output terminals before use for the first time, check whether short circuit exists in the peripheral circuits connected to the MCTC-PES-E1, and check whether the wiring is secure. Failure to comply will result in damage to the MCTC-PES-E1. - Do not change the default settings of the MCTC-PES-E1. Failure to comply may result in damage to the system.
During maintenance	 DANGER	- Do not repair or maintain the MCTC-PES-E1 at power-on. Failure to comply will result in electric shock. - Repair or maintenance of the MCTC-PES-E1 may be performed only by qualified personnel. Failure to comply will result in personal injury or damage to the equipment. - Set and check the parameters again after the MCTC-PES-E1 is replaced. - All the pluggable components must be plugged or removed only after power-off.

1.2 General Precautions

1) Service life

The service life of the MCTC-PES-E1 is 15 years. To ensure normal use and security, you need to replace the product that has been used for approximately 10 years (calculated from delivery) or return them to Monarch for repair and maintenance.

2) Statement on selection of parts

The power supply switches and sensors required by the MCTC-PES-E1 system have been certified by related institutions and must be selected according to the listed models in the manual. Monarch will assume no risk or responsibility if you use the models that are not recommended in the manual.

3) Installation environment requirements

The MCTC-PES-E1 system is installed in the machine room of the escalator or moving walkway. The installation environment must satisfy the requirements described in Table 1-2. Monarch will assume no risk or responsibility if a fault occurs in the system because the installation environment requirements are not satisfied.

Table 1-2 Installation environment requirements

Item	Requirements
Ambient temperature	-20°C to +65°C
Humidity	Less than 95% RH, without condensing
Mounting location	Free from conducting materials, corrosive gas, combustible gas, metal powder, oil dirt and dust
IP level	IP00 (no IP level requirement on board) IP5X (with housing) Note: The MCTC-PES-E1 system has the housing that satisfies IP5X. If you purchase the board only, install the board in the control cabinet or other housing satisfying IP5X.



2

Product Information

Chapter 2 Product Information

2.1 Introduction

The MCTC-PES-E1 is designed to provide multiple security monitoring protections for escalators and moving walkways. It adopts dual-CPU control and judges the signals related to the escalator or moving walkway simultaneously. Once detecting a fault, the two CPUs instruct the safety relays to act and output the fault signal.

Furthermore, the two CPUs monitor each other. If one CPU breaks down or is damaged, the other instructs the safety circuits to act immediately and output the corresponding fault signal. The MCTC-PES-E1 has an LED operation panel, on which you can perform operations such as parameter modification and working status monitoring.

The following figure shows the physical appearance of the MCTC-PES-E1.

Figure 2-1 Physical appearance of the MCTC-PES-E1 (a control board fixed inside the black housing)



2.2 System Components and Functions

2.2.1 System Configuration

The MCTC-PES-E1 is configured with a control board, a protective guard, and a switching-mode power supply (the models are introduced in Table 2-1).

Table 2-1 Models of switching-mode power supply

Component	Quantity	Adapted Model			Remarks
Switching-mode power supply	1	ECU EPR-35-24 (35 W, 24 VDC)	Schneider ABL2REM24015H (35 W, 24 VDC)	Mean Well NES-35-24 (35 W, 24 VDC)	The switching-mode power supply of MCTC-PES-E1 is provided by Monarch.

Note

The switching-mode power supply provides power supply only to the components of the MCTC-PES-E1 (including the control board and the connected sensors and detection switches). It is prohibited to provide power supply to any other devices.

When installing the MCTC-PES-E1, six sensors and seven detection switches are required. The sensor models are designated and sold by Monarch, while the detection switches are decided by the customer. The recommended models of the sensors and detection switches are introduced in Table 2-2.

Monarch will assume no risk or responsibility if the customer uses the models that are not recommended in the manual.

Table 2-2 Models of sensors and detection switches

Accessory	Quantity	Recommended Model		Remarks
Sensor		Schneider	P+F	Specify the required sensor models in your order. Monarch will deliver them (except NBN40-L2-E0-V1) together with the MCTC-PES-E1.
Escalator/Moving walkway speed sensor	2	XS2 18BLNAL2C	NBN8-18GM50-E0-MO	
Handrail speed sensor	2	XS2 12BLNAL2C	NBN4-12GM50-E0-MO	
Step loss detection sensor	2	XS2 30BLNAL2C	NBN15-30GM50-E0-MO NBN40-L2-E0-V1	
Detection Switch		Schmersal	OMRON	Customer option
Motor brake detection switch	2	TS236-02Z TS236-11Z	D4N-4132 D4NA-4131	
Auxiliary brake detection switch	1	TS236-02Z TS236-11Z	D4N-4132	
Floor plate detection switch	4	TS236-02Z TS236-11Z	D4N-4132 D4N-2B64-E1	

2.2.2 System Functions

The MCTC-PES-E1 is designed to ensure safe running of escalator and moving walkway according to the security grading defined in the newest escalator security standard GB16899-2011. It has the following security functions:

- 1) Overspeed protection
- 2) Unintentional reversal protection
- 3) Handrail speed deviation protection
- 4) Service brake and auxiliary brake monitoring protection
- 5) Step or pallet loss protection
- 6) Braking-to-stop distance detection and alarm
- 7) Floor plate state monitoring

■ Working States

The following table describes the states of the MCTC-PES-E1.

Table 2-3 States of the MCTC-PES-E1

State	Description
Startup	The system is powered on and then starts to run properly. In the startup process, the system completes self-check and all relays are in the OFF state.
Setting	You can modify parameters and reset faults using the external operation panel. This mode is active only when the system is in the stop state. When the system is running, you can change nothing using the operation panel.
Stop	The system is not powered on and all relays are in the OFF state.
Fault	The system itself has abnormality, such as self-check failure. In such state, all relays are cut off. You need to power on the system again.
Normal	All functions are normal during running.
Inspection	The system enters the inspection state after the inspection signal becomes active. In such state, the step loss detection function, motor braking detection function and handrail speed detection function are disabled, while the other functions are available.
Manual reset	The system enters the manual reset state after the manual reset signal becomes active. In such state, all security functions are disabled and all relay outputs are in the OFF state.

■ Description of Security Functions

The following table describes the security functions provided by the MCTC-PES-E1.

Table 2-4 Description of security functions

No.	Security Function	Description
1	Over-speed protection 1	Check the speed and acts before the speed exceeds 1.2 times of nominal speed.
2	Over-speed protection 2	Check the speed and act before the speed exceeds 1.4 times of nominal speed.
3	Unintentional reversal protection	Check the unintentional reversal running.
4	Brake monitoring protection	Check the action of the auxiliary brake.
5	Step or pallet loss protection	Check whether the step or pallet is lost.
6	Brake monitoring protection	Check whether the main brake is released.
7	Handrail speed deviation protection	Check the deviation between the handrail speed and the actual step or pallet speed. If the speed deviation is greater than -15% and continues for 15s, the system performs protection.
8	Braking-to-stop distance detection and alarm	Check the braking-to-stop distance. If the distance exceeds 1.2 times of the allowable maximum value, the system performs protection.
9	Floor plate detection switch	Check whether the floor plate is opened or removed.

■ Security Measures After Security Functions Enabled

The following table describes the security measures after the security functions become enabled.

Table 2-5 Security measures after the security functions enabled

No.	Description of Security Function	Security Circuit Open	Power off the Supply of Auxiliary Brake	Manual Reset
1	Check the speed and acts before the speed exceeds 1.2 times of nominal speed.	Yes	No	Yes
2	Check the speed and acts before the speed exceeds 1.4 times of nominal speed.	Yes	Yes	Yes
3	Check the unintentional reversal running.	Yes	Yes	Yes
4	Check the action of the auxiliary brake.	Yes	No	No
5	Check whether the step or pallet is lost.	Yes	No	Yes
6	Check whether the main brake is released.	Yes	No	Yes
7	Check the deviation between the handrail speed and the actual step or pallet speed. If the speed deviation is greater than -15% and the duration exceeds 15s, it performs protection.	Yes	No	No
8	Check the braking-to-stop distance. If the distance exceeds 1.2 times of the allowable maximum value, it performs protection.	Yes	No	Yes
9	Check whether the floor plate is opened/ removed.	Yes (see the "Note" below.)	No	No

Note

The security circuit of the escalator or moving walkway is cut off after the floor plate is opened or removed. After the system enters the inspection state, the fault is reset automatically.



3

Mechanical and Electrical Installation

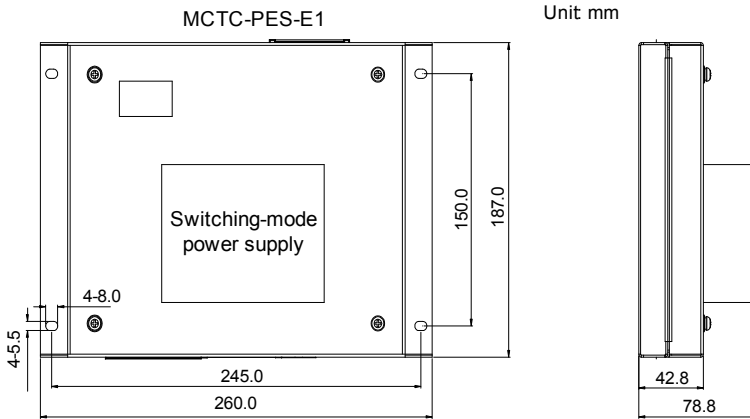
Chapter 3 Mechanical and Electrical Installation

3.1 Mechanical Installation

3.1.1 Installation Dimensions

The following figure shows the dimensions of the MCTC-PES-E1 system.

Figure 3-1 Dimensions of the MCTC-PES-E1 system



Only the control board (without the protective guard) is required in some applications. The following figures show the physical appearance and installation dimensions of the control board.

Figure 3-2 Physical appearance of the control board

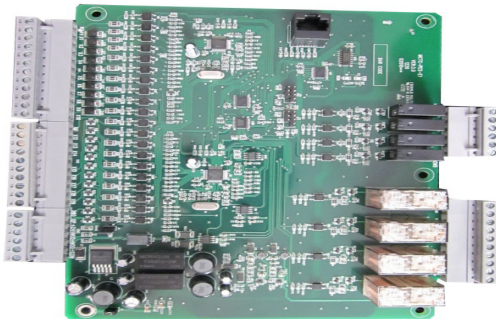
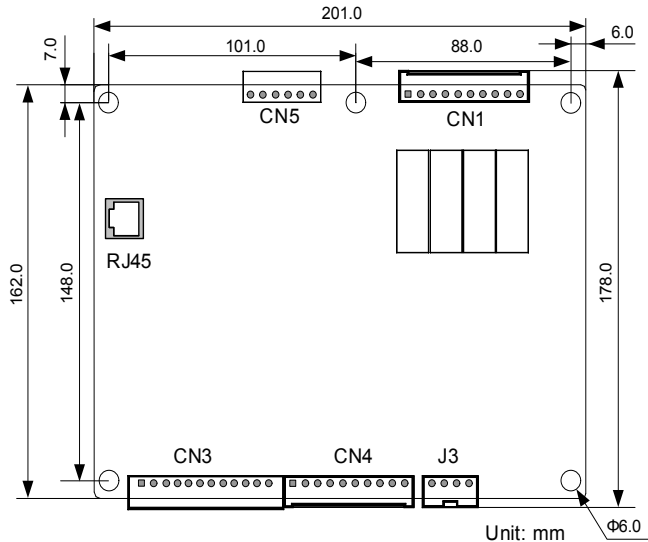


Figure 3-3 Installation dimensions of the control board



3.1.2 Installation Method and Precautions

To install the MCTC-PES-E1 system, do as follows:

1. Process the installation side in the main control cabinet. For the processing dimensions, refer to Figure 3-1.
2. Mount the MCTC-PES-E1 on the installation side and fix it with screws.

If only the control board (without the protective guard) is required, do as follows:

1. Process the installation side in the main control cabinet. For the processing dimensions, refer to Figure 3-3.
2. Mount the control board on the installation side and fix it with screws.

Observe the following precautions during installation:

- 1) The switching-mode power supply has been fixed on the protective guard of the MCTC-PES-E1 system upon delivery.
- 2) The screws for fixing the protective guard are not delivered together with the MCTC-PES-E1 and you need to prepare them yourself. When selecting screws, consider the plate thickness of the installation side, the thickness of the protective guard and the dimension of the hole in Figure 3-1.
- 3) The MCTC-PES-E1 system is installed in the main control cabinet of the machine room of the escalator or moving walkway. Keep the installation room clean, well ventilated and dust & moisture proofing. Especially when only the control board without the protective guard is installed, take other dust & moisture proofing measures.

3.2 Installation of Sensors

3.2.1 Technical Data of Sensors

The following table describes the technical data of the sensors.

Table 3-1 Technical data of Schneider sensors

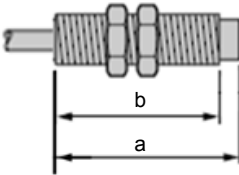
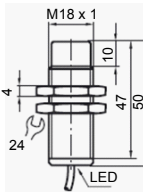
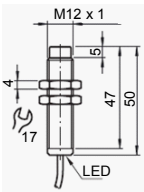
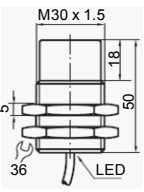
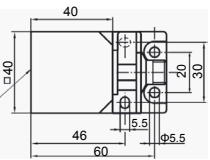
Brand	Schneider		
Sensor	Escalator/Moving walkway speed sensor	Handrail speed sensor	Step loss detection sensor
Model	XS2 18BLNAL2C	XS2 12BLNAL2C	XS2 30BLNAL2C
Diameter	18 mm	12 mm	30 mm
Sensing Distance	≤ 8 mm	≤ 4 mm	≤ 15 mm
Length	50.6 mm	41.3 mm	50.6 mm
Power Supply (DC)	12–24 V	12–24 V	12–24 V
Output	NPN	NPN	NPN
Definition of Cables	Brown (BN): positive Blue (BU): negative Black (BK): output signal	Brown (BN): positive Blue (BU): negative Black (BK): output signal	Brown (BN): positive Blue (BU): negative Black (BK): output signal
Appearance			
Dimension	a = 50.6 mm b = 40.4 mm	a = 50.6 mm b = 34.1 mm	a = 50.6 mm b = 35.4 mm
			

Table 3-2 Technical data of the P+F sensors

Brand	P+F			
Sensor	Escalator/ Moving walkway speed sensor	Handrail speed sensor	Step loss detection sensor	
Model	NBN8-18GM50- E0-MO	NBN4-12GM50- E0-MO	NBN15- 30GM50-E0-MO	NBN40-L2-E0-V1
Diameter	18 mm	12 mm	30 mm	Square (see the dimension)
Sensing Distance	≤ 8 mm	≤ 4 mm	≤ 15 mm	≤ 40 mm
Length	50 mm	50 mm	50 mm	See the dimension.
Power Supply (DC)	10–30 V	10–30 V	10–30 V	10–30 V
Output	NPN	NPN	NPN	NPN
Definition of Cables	Brown (BN): positive Blue (BU): negative Black (BK): output signal	Brown (BN): positive Blue (BU): negative Black (BK): output signal	Brown (BN): positive Blue (BU): negative Black (BK): output signal	Brown (BN): positive Blue (BU): negative Black (BK): output signal Do not connect the white cable.
Appearance				
Dimension				

Note

- All the sensor models except NBN40-L2-E0-V1 are obtainable in Monarch.
- The principle on selecting step or pallet loss sensor is as follow:
 - For iron step or pallet, select XS2 30BLNAL2C or NBN15-30GM50-E0-MO.
 - For aluminum step or pallet, it is suggested to select NBN40-L2-E0-V1 (P+F), whose detection distance is long and signal is reliable.

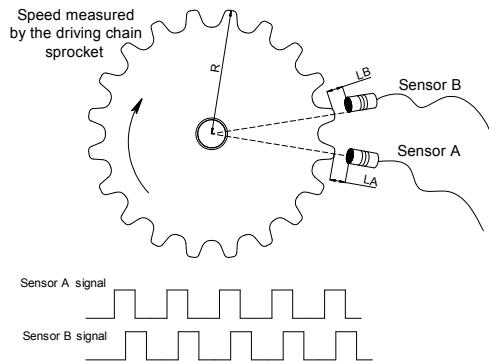
3.2.2 Installing the Speed Sensor and Running Direction Sensor for Escalator/Moving Walkway

■ Installation

- 1) Align the center of the sensing face of one sensor with the center of one tooth on the driving chain sprocket and align the sensing edge of the other sensor with the center of the neighboring tooth on the driving chain sprocket, as shown in the following figure.
- 2) Installation distance $3 \text{ mm} \leq LA = LB \leq 8 \text{ mm}$

The following figure shows the installation of the escalator/moving walkway speed sensor and running direction sensor.

Figure 3-4 Installation of the escalator/moving walkway speed sensor and running direction sensor



■ Detection Principle

- 1) Overspeed protection function

By measuring the speed of the driving chain sprocket through sensor A and sensor B, the MCTC-PES-E1 system judges whether the running speed of the escalator/moving walkway exceeds the allowable speed. If yes, the MCTC-PES-E1 system will perform overspeed protection.

When the driving chain sprocket rotates, every time a sensor detects a tooth, the sensor sends a pulse. The MCTC-PES-E1 system calculates the running speed of the escalator/moving walkway based on the time interval of detected pulses.

Sensor A and sensor B are redundant speed measuring channels for each other. By setting a certain pulse cycle or frequency threshold, 1.2 times or 1.4 times overspeed can be detected.

- 2) Unintentional reversal protection

By fixing the relative positions of the two sensors correctly, make the phase of sensor A in advance of the phase of sensor B and ensure that the two sensors have overlapping pulse part. In this case, the MCTC-PES-E1 system detects the logic sequence of the two sensors and judges the actual running direction of the escalator/moving walkway through the logic sequence, preventing reversal running.

3.2.3 Installing the Handrail Speed Sensor

■ Installation

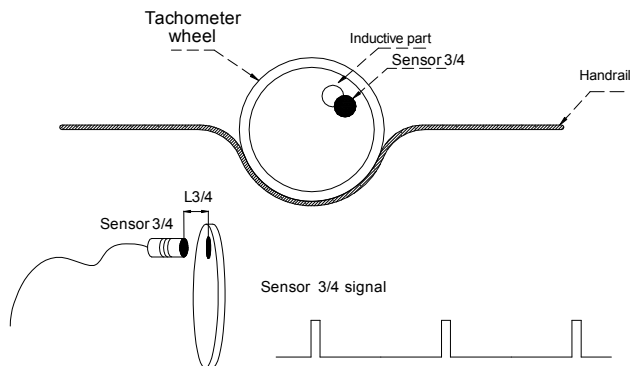
- 1) Align the sensor with the inductive part on the tachometer wheel.

If the tachometer wheel is made of plastic, use a ferric device as the inductive part. If the tachometer wheel is ferric, drill a hole as the inductive part in the wheel. The section size of the inductive part should be approximate to that of the sensing face of the sensor.

- 2) Installation distance: $1\text{ mm} \leq L3 = L4 \leq 4\text{ mm}$

The following figure shows the installation position of the handrail speed sensor.

Figure 3-5 Installation position of the handrail speed sensor



■ Detection Principle

Sensor 3/4 is used to measure the left/right handrail speed. The tachometer wheel is driven by the handrail to rotate and its linear speed is basically consistent with the handrail speed. An inductive part is set on the tachometer wheel to mount sensor 3/4 on a fixed mechanical structure with the sensing face toward the inductive part.

When the tachometer wheel follows the handrail to rotate, sensor 3/4 outputs pulses, as shown in Figure 3-5. A pulse is output every time the tachometer wheel rotates a revolution.

With the pulses and radius of the wheel, the system can measure the rotational speed of the tachometer wheel. Then the system calculates the handrail speed and compares it with the escalator/moving walkway speed. If the handrail speed is lower than 85% of the corresponding escalator/moving walkway speed and the handrail underspeed continues for 15s, the system cuts off the power supply to the safety circuit of the escalator/moving walkway to stop the running, implementing protection.

3.2.4 Installing the Step/Pallet Loss Sensor

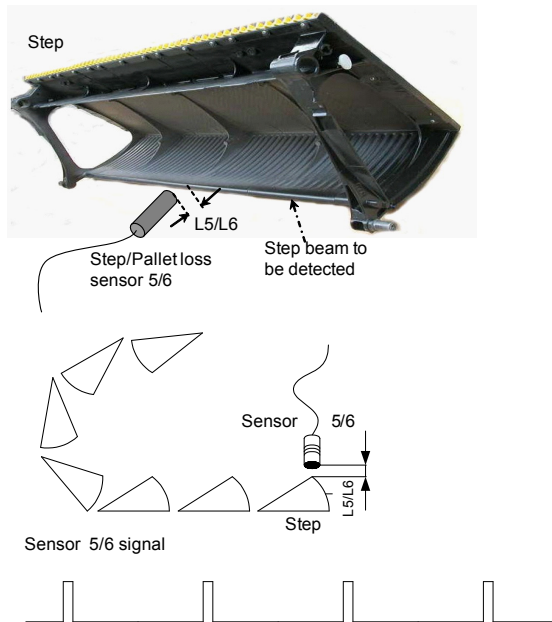
1) Installation

For aluminum step or pallet, it is suggested to select NBN40-L2-E0-V1 (P+F), whose detection distance is long and signal is reliable. Each step/pallet loss sensor is required for the top and bottom machine rooms respectively. Install the sensor with the sensing face toward the step beam to be detected.

2) Installation distance: $5\text{ mm} \leq L5 = L6 \leq 15\text{ mm}$

The following figure shows the installation position of sensor (5/6) for the escalator.

Figure 3-6 Installation position of the step/pallet loss sensor for escalator



3) Detection Principle

Each sensor (5/6) is installed in the return/driving station in the top/bottom machine room, respectively to detect step loss. Based on signals of speed sensor A/B for the motor, the system calculates the number of pulses of sensor A/B between two adjacent pulses of sensor 5/6 to judge whether step/pallet loss occurs.

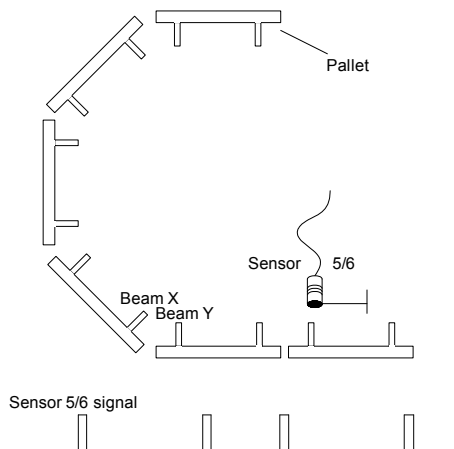
When a step passes, sensor 5/6 receives the signal and outputs a pulse. Set the time interval between two adjacent pulses of sensor 5/6 as T and the pulse counting of sensor A/B within T as X . In the case of no step loss, the X value is within a certain threshold regardless of the escalator speed. If X exceeds the threshold, the system judges that step loss occurs. Then the escalator stops running immediately and enters the safety state.

Note

If the detection principle is not suitable to the use habit of escalator manufacturers, step loss can be monitored by using sensor 5/6 to detect the step auxiliary wheel. The prerequisite is to ensure that the auxiliary wheel is lost while the step gets lost in any way.

To detect whether a pallet of the moving walkway is lost, install sensor 5/6 in the machine room, as shown in the following figure.

Figure 3-7 Installation position of the step/pallet loss sensor for moving walkway

**⚠ WARNING**

Besides the MCTC-PES-E1 and the six sensors, you need to prepare the following two devices:

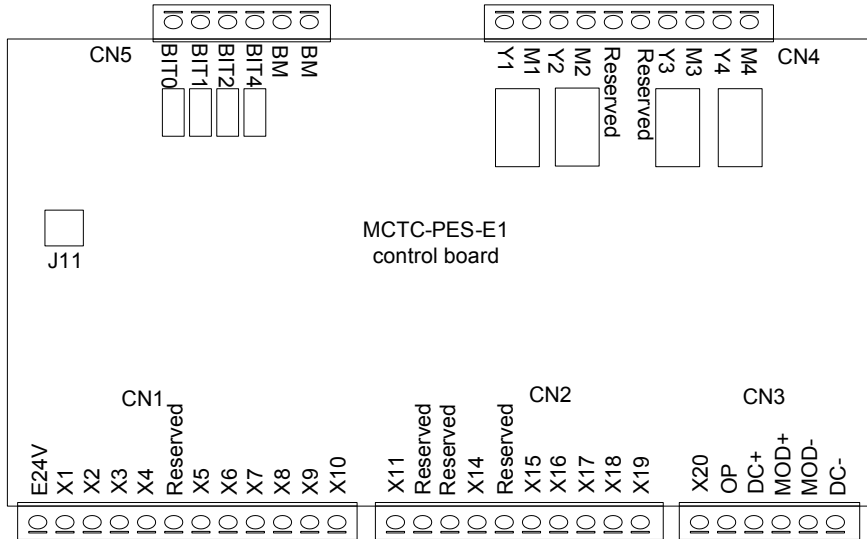
- Relay at front end of signal X1 on the MCTC-PES-E1 board
- Switch for manual fault reset

3.3 Electrical Installation

3.3.1 Description of System Terminals

The following figure shows the layout of the system terminals.

Figure 3-8 Layout of the system terminals



The following table describes the terminals of the MCTC-PES-E1.

Table 3-3 Description of terminals of the MCTC-PES-E1

Type	Terminal Symbol	Terminal Name	Function Description	Remark
Power/Communication terminal (CN3)	DC+/DC-	24 V power supply	The + 24 VDC power is provided by specialized switching-mode power supply.	The switching-mode power supply complies with the EN60950 standard.
	MOD+/MOD-	Modbus communication	It communicates with Inovance NICE2000 integrated escalator controller. When a safety fault occurs, the NICE2000 displays the fault code.	The shielded twisted cable is suggested.

Type	Terminal Symbol	Terminal Name	Function Description	Remark
Digital input terminal (CN1/2)	E24V	External +24 V power supply	It is connected to the input terminal 24V of the control system when X1 to X4 are connected with corresponding signals of the control system in parallel.	The switching-mode power supply complies with the EN60950 standard.
	X1	Normally open (NO) inspection signal	They are connected to the inspection signal input terminals of the control system.	1) Optical coupling isolation, low level input active 2) Input impedance: 3.3 kΩ 3) Input voltage range: 0–30 V
	X2	Normally closed (NC) inspection signal	When X1/ X2 becomes ON, the related functions of the MCTC-PES-E1 are disabled according to GB standard.	
	X3	Up signal	It receives the up signal of the control system and is edge-triggered.	
	X4	Down signal	It receives the down signal of the control system and is edge-triggered.	
	X5	Service brake signal	It is connected to the auxiliary contact of the service brake.	
	X6	Service brake action detection 1	It is connected to the action switch of the service brake.	
	X7	Service brake action detection 2	It is used in the dual-motor application. It is shorted with X6 in the single-motor application.	
	X8	Auxiliary brake signal	It is connected to the auxiliary contact of the auxiliary brake.	
	X9	Auxiliary brake action detection	It is connected to the action switch of the auxiliary brake.	
	X10	Floor plate switch 1	It is connected to the floor plate switch. This signal must be active during normal running and can be inactive during inspection.	
	X11	Floor plate switch 2	It has the same function as the X10.	
	X12	Reserved		
	X13	Reserved		
	X14	Manual reset signal	It is used for manual fault reset. The input becomes active after the terminal keeps ON for more than 2s and then keeps OFF for more than 2s.	

Type	Terminal Symbol	Terminal Name	Function Description	Remark
Digital input terminal (CN1/2)	X15	Escalator speed detection phase A signal	They are used for input of escalator speed detection pulses The MCTC-PES-E1 detects the overspeed fault based on the frequency or cycle of two pulses, and detects the unintentional reversal fault based on the pulse phase.	1) Optical coupling isolation, low level input active 2) Input impedance: 3.3 kΩ 3) Input voltage range: 0–30 V
	X16	Escalator speed detection phase B signal		
	X17	Up step loss detection	It detects the A/B pulses. If the pulse counting exceeds the threshold, the MCTC-PES-E1 judges that it is step loss.	
	X18	Down step loss detection	It uses the same detection mode as X17.	
	X19	Left handrail speed detection	If the MCTC-PES-E1 detects that the handrail speed is 15% lower than the actual speed of the step, pallet or step chain and the duration exceeds 15s, it reports the handrail underspeed fault.	
	X20	Right handrail speed detection		
	OP	X15 to X20 polarity selection	When X15 to X20 are connected to sensors of low level active, OP is shorted to DC+. When X15 to X20 are connected to sensors of high level active, OP is shorted to DC-.	Shorting OP and DC+/DC- is effective for X15 to X20 only.
Relay output terminal (CN4)	Y1/M1	Fault output	It becomes OFF when a fault occurs. It is connected to the system safety circuit with Y2/M2.	Rated voltage: 250 VAC Rated current: 6 A
	Y2/M2		It becomes OFF when a fault occurs. It is connected to the system safety circuit with Y1/M1.	
	Y3/M3	Auxiliary brake output	It becomes OFF when 1.4 times overspeed or unintentional reversal running occurs. It is connected with Y4/M4 to control the auxiliary brake.	
	Y4/M4		It becomes OFF when 1.4 times overspeed or unintentional reversal running occurs. It is connected with Y3/M3 to control the auxiliary brake.	

Type	Terminal Symbol	Terminal Name	Function Description	Remark
Power/ Communication terminal (CN3)	BIT0	Fault binary code	It outputs the current fault code using the 8421 code and can be connected to the external LED display. It supports both common anode LED display and common cathode LED display.	A common relay is used.
	BIT1			
	BIT2			
	BIT3			
	BM	Common terminal		
Interface for operation panel (J11)	RJ45	Interface for the operation panel	It is connected to the operation panel for modify ing parameters or viewing the status.	-

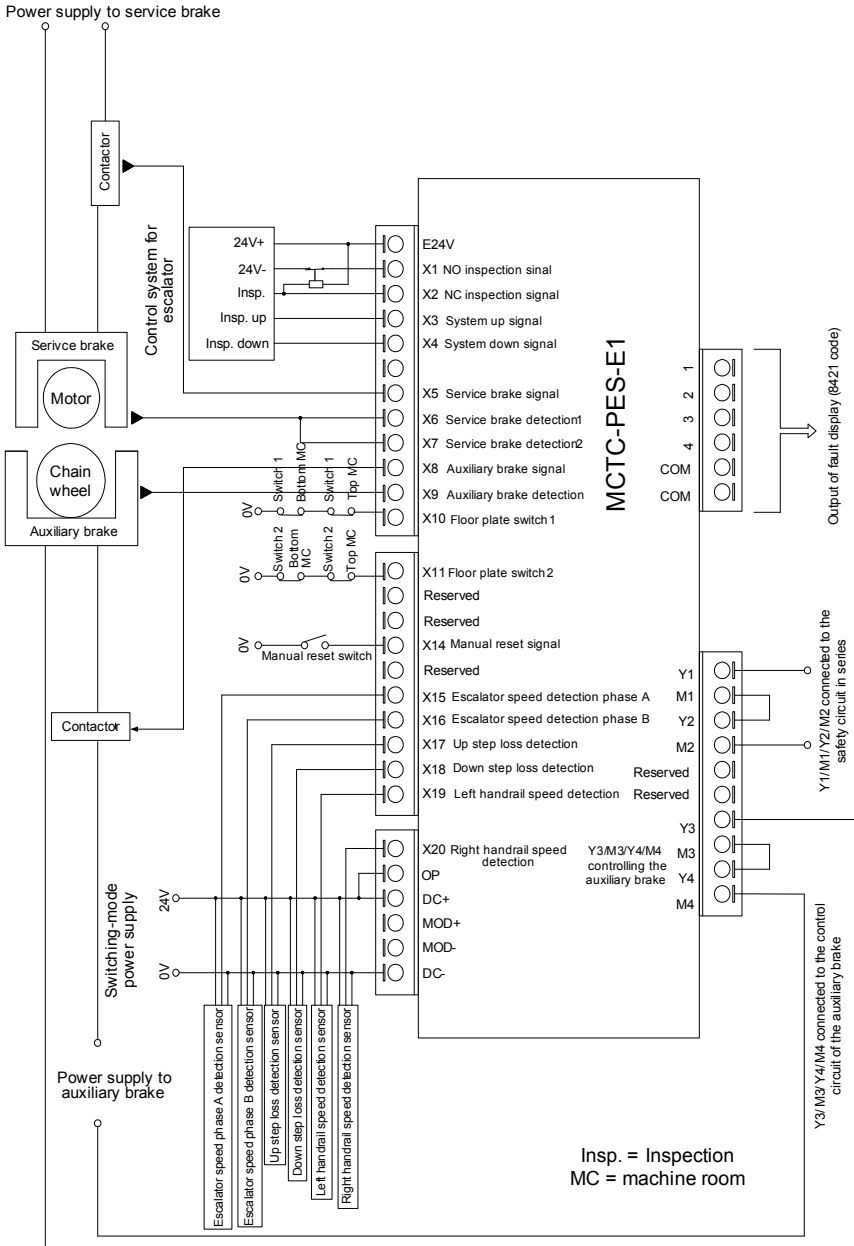
Note

- If there is no auxiliary brake in the system, short X8 and X9 to DC-.
 - If the floor plate switches in the system have the compulsory break-off function, the related function of this system can be disabled. In this case, short X10 and X11 to DC-.
-
-

3.3.3 Wiring of the MCTC-PES-E1

The following figure shows wiring of the MCTC-PES-E1 system.

Figure 3-9 Wiring of the MCTC-PES-E1 system



The precautions on electrical wiring are as follows:

- 1) All input signals are low level active.
- 2) E24V is connected to the 24 V power supply of the escalator control system. DC+/DC- is connected to the switching-mode power supply of the MCTC-PES-E1. The X1 to X4 terminals are connected to the 24 V power supply of the escalator control system and cannot use the power supply of the MCTC-PES-E1.
- 3) X1, X3 to X9 and X14 are NO signals by default, while X2, X10 and X11 are NC signals by default. The NO/NC feature of X1 to X5, X8 and X14 cannot be modified.
- 4) Because X1 is NO and X2 is NC, you need to install an NC relay before X1. The wiring method is shown in Figure 3-8. If X1 and X2 open or close simultaneously, the MCTC-PES-E1 reports a fault.



WARNING

The inspection signal connecting to the relay before X1 and that connecting to X2 must be obtained separately from the inspection switch and cannot share one cable.

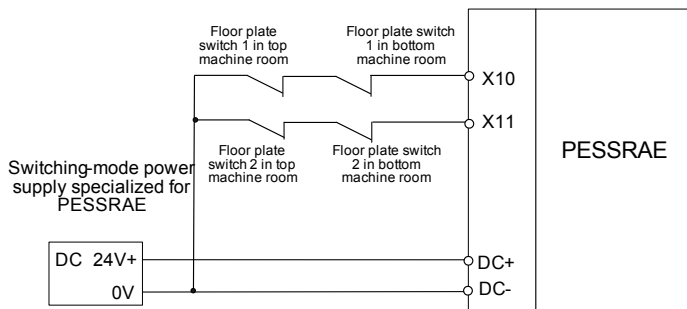
If there is no auxiliary brake, short X8 and X9 to 0V. If the service brake uses an inspection switch, short X6 to X7.

- 5) If the floor plate switches have the compulsory break-off function, the function of X10 and X11 can be disabled. In this case, X10 and X11 must be active, which can be implemented by shorting them or modifying their NO/NC feature.
- 6) OP is active only for X15 to X20. For low-level active signals (NPN output), short OP to 24V. Otherwise, short OP to 0V.

3.3.3 Connecting the Floor Plate Switches

The system can detect opening or removal of the floor plate. If the floor plate detection function is enabled, a total of four floor plate switches are required. The top machine room and bottom machine room respectively need to install two detection switches. The following figure shows connection between the four detection switches and the system.

Figure 3-10 Connection of the four floor plate switches



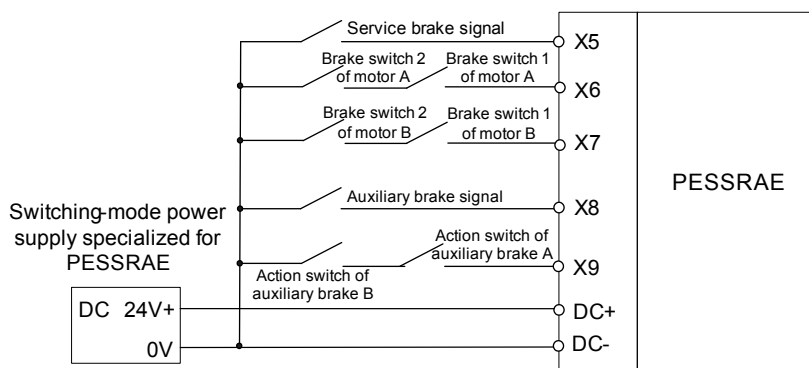
As shown in Figure 3-10, the two switches in the top machine room must be connected to the two switches in the bottom machine room in series. The two signals are respectively connected to X10 and X11. Thus, if any floor plate is removed, both signals have a change.

3.3.4 Connecting Brake Detection Switches in Dual-Motor Application

In the application of dual motors and dual brakes, to satisfy China's GB standard, obey the following instructions:

- 1) Each of the dual motors has a service brake. The two brakes can be controlled by the same contactor. The contactor is connected to the NO auxiliary contact of the service brake signal X5
- 2) The two auxiliary brakes can also be controlled by a contactor. The contactor is connected to the NO auxiliary contact of the auxiliary brake signal X8.
- 3) When the service brake is applied, its action switches must be in the OFF state and be connected according to Figure 3-11. That is, the two action switches of the brake for motor A are connected in series and used as the input of X6. The two action switches of the brake for motor B are connected in series and used as the input of X7.
- 4) When the auxiliary brake is applied, its action switches must be in the OFF state and be connected according to Figure 3-11. That is, the two action switches of auxiliary brake A and auxiliary brake B are connected in series and used as the input of X9.

Figure 3-11 Connecting brake detection switches in the dual-motor application





4

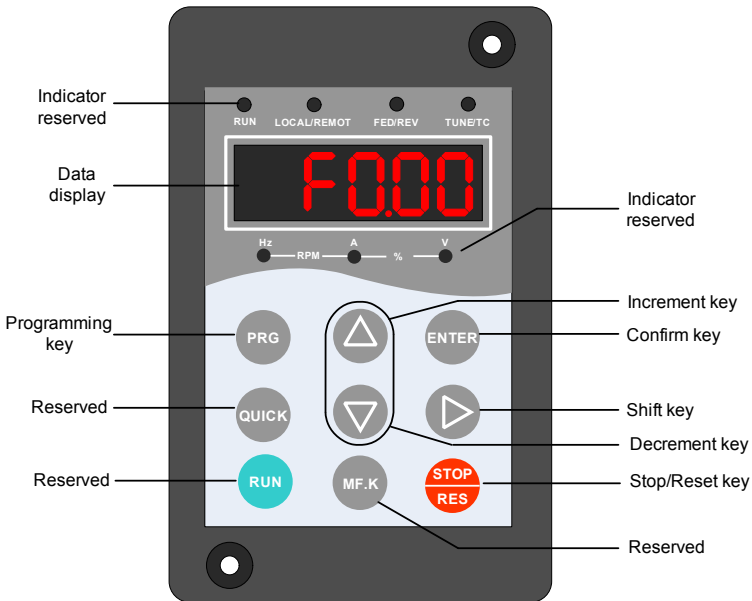
System Operation and Commissioning

Chapter 4 System Operation and Commissioning

4.1 Operation Panel

The MCTC-PES-E1 system has a built-in LED operation panel on it. You can modify the parameters, monitor the working status and view the fault information by operating the operation panel, as shown in the following figure.

Figure 4-1 Diagram of the operation panel



4.1.1 Description of Keys on the Operation Panel

Table 4-1 Description of keys on the operation panel

Key	Name	Function
	Programming	Enter or exit Level I menu.
	Confirm	Enter the menu interfaces level by level, and confirm the parameter setting.
	Increment	Increase data or function code.
	Decrement	Decrease data or function code.

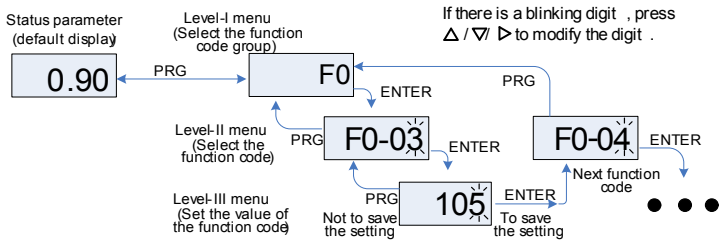
Key	Name	Function
	Shift	Select the digit to be modified when modifying parameters.
	Reserved	Reserved
	Stop/Reset	Perform the reset operation when the system is in the fault state.
	Reserved	Reserved
	Reserved	Reserved

4.2 Viewing and Modifying Function Codes

The operation panel of the MCTC-PES-E1 adopts three-level menu.

The three-level menu consists of function code group (Level I), function code (Level II), and function code setting value (level III), as shown in the following figure.

Figure 4-2 Operation procedure on the operation panel

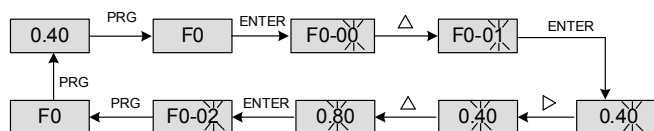


You can return to Level II menu from Level III menu by pressing or . The differences are as follows:

- After you press , the system saves the parameter setting first, and then goes back to Level II menu and shifts to the next function code.
- After you press , the system does not save the parameter setting, but directly returns to Level II menu and remains at the current function code.

Here is an example of changing the value of F0-01 from 0.40 m/s to 0.80 m/s.

Figure 4-3 Example of changing the parameter value



In Level III menu, if the parameter has no blinking digit, it means that the parameter cannot be modified. This may be because:

- Such a function code is only readable, such as, actually detected parameter and running record parameter.
- Such a function code cannot be modified in the running state and can be changed only at stop.

4.3 Function Code Description and Applications

The symbols in the function code tables are described as follows:

★: The parameter can be viewed but cannot be modified when the system is in the running state. It can be modified when the system is in the stop state.

•: The parameter is the actually measured value and cannot be modified.

In commissioning, you only need to adjust the parameters in group F0. Group F1 contains state parameters, group F2 contains fault parameters, FF contains factory parameters and group FP contains the specialized application parameters.

Group F0: Basic Parameters

Function Code	Parameter Name	Setting Range	Default	Unit	Property
F0-00	System type	0: Escalator 1: Moving walkway	0	-	★

It is used to set the system type and defines the upper limit of nominal speed (F0-01).

Function Code	Parameter Name	Setting Range	Default	Unit	Property
F0-01	Nominal speed	0.30–0.90	0.50	m/s	★

It indicates the nominal speed when the escalator/moving walkway is in the no-load state.

Function Code	Parameter Name	Setting Range	Default	Unit	Property
F0-02	Radius of driving chain sprocket	0.30–0.90	0.33	m	★

It indicates the radius of the driving chain sprocket. The linear speed of the driving chain sprocket calculated based on the radius is the step running speed.

Function Code	Parameter Name	Setting Range	Default	Unit	Property
F0-03	PPR of driving chain sprocket	1–200	65	-	★

It is used to set the pulses per revolution (PPR) of the driving chain sprocket, namely, the number of teeth of the driving chain sprocket.

Function Code	Parameter Name	Setting Range	Default	Unit	Property
F0-04	Maximum braking-to-stop distance	0.20–1.69	0.5	m	★

The upper limit and lower limit of this parameter are controlled by F0-01 (Nominal speed). If the system detects that the braking-to-stop distance exceeds 1.2 times of the setting of F0-04, the system reports the braking-to-stop over-distance fault.

Function Code	Parameter Name	Setting Range	Default	Unit	Property
F0-05	Handrail pulse time interval at nominal speed	0.01–5.00	0.68	s	★

It is used to set the time interval between handrail pulses at nominal speed. The system detects whether the handrail speed is consistent with the escalator speed based on the nominal speed, the setting of F0-05 and the actually detected escalator speed.

For example, the nominal speed is 0.5 m/s and F0-05 is set to 0.10s. When the actually detected escalator speed is 0.4 m/s, the time interval between handrail signals should be $0.5 \times 0.1 / 0.4 = 0.125\text{s}$. If the actually detected time interval between handrail signals is longer than 0.125s, it indicates that the handrail is slow. If the detected handrail speed remains 15% slower than the escalator speed for 15s, the system reports the related fault.

Function Code	Parameter Name	Setting Range	Default	Unit	Property
F0-06	Upper limit of phase A/B pulses between two step signals	-	0	-	★

It is used to set the upper limit of phase A/B pulses between two step signals. For every step the escalator turns, the number of teeth of the driving chain sprocket counted by the sensor is fixed.

After you set F0-06 to 0 first, the system automatically calculates the number of phase A/B pulses between two step signals and saves it. Then you can set the upper limit of phase A/B pulses between two step signals according to the following principle:

If the automatic calculation value is A, it is suggested to set 2 times value A limitation for F0-06.

Function Code	Parameter Name	Setting Range	Default	Unit	Property
F0-07	Lower limit of phase A/B pulses between two step signals	0 to F0-06	0	-	★

It is used to set the lower limit of phase A/B pulses between two step signals and should be a little smaller than the automatic calculation value of the system.

- For the escalator, it is mainly used to filter the incorrect signal that may arise (for example, interference) between two adjacent step signals.
- For the moving walkway, the value of F0-07 should be larger than the number of phase A/B pulses between the detected beams of two adjacent pallets (for example, beams X and Y in Figure 3-7). Otherwise, the system will report fault Err12.

Function Code	Parameter Name	Setting Range	Default	Unit	Property
F0-08	Auxiliary function selection	0–65535	0	-	★

The auxiliary functions of the system are described in the following table.

Table 4-2 Auxiliary functions indicated by bits of F0-08

Bit	Description
Bit0	0: Escalator running not available when the auxiliary brake is applied 1: Escalator running available only in the upward direction when the auxiliary brake is applied For some escalators, the auxiliary brake has the braking effect only in the downward direction and can be released only after the escalator runs upward for a certain distance. In this case, set Bit0 to 1. The running in the upward direction can last only 10s each time. If the running times out, the MCTC-PES-E1 disconnects outputs, forces the escalator to stop running and reports fault Err10. After the fault is reset, the escalator can run upward for 10s again.
Bit1	0: Auxiliary brake available 1: Auxiliary brake unavailable Not all escalators must install the auxiliary brake. If Bit1 is set to 1, the MCTC-PES-E1 disconnects only Y3 and Y4 (the two relays stop the output).
Bit2	0: Step loss not detected during inspection 1: Step loss detected during inspection
Bit3	0: Handrail underspeed not detected during inspection 1: Handrail underspeed detected during inspection
Bit4	0: Brake not detected during inspection 1: Brake detected during inspection
Bit5	0: Detection of handrail speed pulse frequency too high enabled 1: Detection of handrail speed pulse frequency too high disabled
Bit6	Reserved
Bit7	0: Feedback signal detected after brake release 1: Feedback signal not detected after brake release

Note

The signals input by X10 and X11 are not detected in the system inspection state.

Group F1: State Parameters

Function Code	Parameter Name	Setting Range	Default	Unit	Property
F1-00	Phase A escalator speed	-	0	m/s	●
F1-01	Phase B escalator speed	-	0	m/s	●

They indicate the escalator speed respectively, detected by sensor A and sensor B.

Function Code	Parameter Name	Setting Range	Default	Unit	Property
F1-02	Number of phase A pulses/ cycles for escalator speed	-	0	-	●
F1-03	Number of phase B pulses/ cycles for escalator speed	-	0	-	●

If the number of phase A/B pulses per second at nominal speed (F0-01) is equal to or larger than 30, F1-02 and F1-03 display the actually detected number of phase A/B pulses. If their value is smaller than 30, the phase A/B cycles is displayed by a 3-digit decimal in the unit of second.

Function Code	Parameter Name	Setting Range	Default	Unit	Property
F1-04	Current running direction	-	0	-	●

It displays the escalator speed signal state. The 5-digit display is abcd from left to right.

Display	Description
a	0 indicates that the direction is not given. 1 indicates that the startup command in upward direction is given. 2 indicates that the startup command in downward direction is given.
b	0 indicates that the feedback signal is abnormal. 1 indicates that the feedback running direction is upward. 2 indicates that the feedback running direction is downward.
c, d	Reserved
e	It represents the orthogonality. 0 indicates the worst, and 9 indicates the best. Ensure the value above 5 by adjusting the position of the two speed sensors.

The parameter has three normal values: 00000, 21001 and 10110.

Function Code	Parameter Name	Setting Range	Default	Unit	Property
F1-05	Braking-to-stop distance after stop	-	0	m	●

It indicates the detected braking-to-stop distance after stop.

Function Code	Parameter Name	Setting Range	Default	Unit	Property
F1-06	Pulse time interval of left handrail	-	0	s	●
F1-07	Pulse time interval of right handrail	-	0	s	●

They indicate the actually detected cycles of left and right handrail signals (no display during inspection).

Function Code	Parameter Name	Setting Range	Default	Unit	Property
F1-08	Number of phase A pulses between two up step signals	-	0	-	●
F1-09	Number of phase B pulses between two up step signals	-	0	-	●
F1-10	Number of phase A pulses between two down step signals	-	0	-	●
F1-11	Number of phase B pulses between down step signals	-	0	-	●

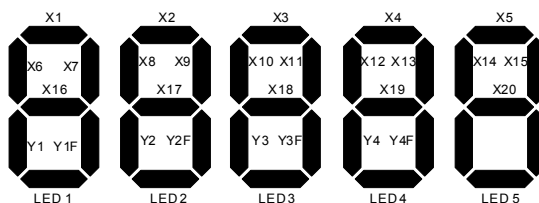
They indicate the actually detected number of phase A/B pulses between two step signals (no display during inspection).

Function Code	Parameter Name	Setting Range	Default	Unit	Property
F1-12	States of I/O terminals	-	0	-	●
F1-13	States of I/O terminal functions	-	0	-	●

F1-12 indicates the states of the I/O terminals. F1-13 indicates the states of the I/O terminal functions.

The states are displayed by the following 5-digit LEDs.

Figure 4-4 LED indicating the states of the I/O terminals and their functions



If OP is shorted to DC+, input terminals X15 to X20 are low level active. If OP is shorted to DC-, X15 to X20 are high level active. The other input terminals are low level active and become ON when they are shorted to 0V.

The function of a terminal indicated by an LED segment in the preceding figure may be valid when this segment is ON or OFF. Whether this function is valid at segment ON or OFF is decided by the NO/NC feature of the terminal. Y1F to Y4F are detection signals and indicate the action state of the detected relays Y1 to Y4. In normal state, Y1F to Y4F are ON or FF with Y1 to Y4 simultaneously.

Function Code	Parameter Name	Setting Range	Default	Unit	Property
F1-14	Number of pulses per second at nominal speed	-	0	-	●

It indicates the number of phase A/B pulses per second at nominal speed, which is calculated based on parameters in group F0.

Function Code	Parameter Name	Setting Range	Default	Unit	Property
F1-15	Number of pulses for 1.2 times of braking-to-stop distance	-	0	-	●

It indicates the number of phase A/B pulses for 1.2 times of braking-to-stop distance, which is calculated based on parameters in group F0.

Function Code	Parameter Name	Setting Range	Default	Unit	Property
F1-16	Number of pulses between two adjacent step signals	-	0	-	●

It indicates the number of phase A/B pulses between two adjacent step signals, which is calculated based on parameters in group F0 and the escalator step of 40 cm wide.

Function Code	Parameter Name	Setting Range	Default	Unit	Property
F1-17	ER-00	-	0	-	●

It indicates the faults that have occurred and need to be reset manually. Each fault is allocated with a bit. The corresponding relationship is as follows:

Bit0: Err01; Bit1: Err02;Bit15: Err16

After the system reports the fault, the corresponding bit will become valid and is saved in F1-17 after being converted to a decimal number.

Note

This parameter records only the faults that need manual reset. After the fault is reset manually, the fault record is cleared.

Function Code	Parameter Name	Setting Range	Default	Unit	Property
F1-18	Err Flag	-	0	-	●

It indicates the faults that have occurred. Besides the faults that need manual reset, it records the faults that can be reset at power failure.

Function Code	Parameter Name	Setting Range	Default	Unit	Property
F1-19	Current MCTC-PES-E1 state	-	0	-	●

The unit's digit indicates the auxiliary chip state, and the ten's digit indicates the main chip state. On normal condition, the main chip and auxiliary chip are in the same state. The chips have a total of four states, indicated by 3, 4, 5 and 6, whose meaning is as follows:

- 3: MCTC-PES-E1 faulty
- 4: MCTC-PES-E1 in normal working state
- 5: Inspection state
- 6: fault reset state

Group F2: Fault Information

Function Code	Parameter Name	Setting Range	Default	Unit	Property
F2-00	1st fault code	-	-	-	●
F2-01	1st fault sub-code	-	-	-	●
F2-02	2nd fault code	-	-	-	●
F2-03	2nd fault Sub-code	-	-	-	●
F2-04	3rd fault code	-	-	-	●
F2-05	3rd fault sub-code	-	-	-	●
F2-06	4th fault code	-	-	-	●
F2-07	4th fault sub-code	-	-	-	●
F2-08	Latest fault code	-	-	-	●
F2-09	latest fault sub-code	-	-	-	●
F2-10	Nominal speed at latest fault	-	-	-	●
F2-11	Radius of driving chain sprocket at latest fault	-	-	-	●
F2-12	PPR of driving chain sprocket latest fault	-	-	-	●
F2-13	Maximum braking-to-stop distance at latest fault	-	-	-	●
F2-14	Handrail pulse time interval at nominal speed at latest fault	-	-	-	●
F2-15	Upper limit of phase A/B pulses for step signals at nominal speed at latest fault	-	-	-	●
F2-16	Lower limit of phase A/B pulses for step signals at nominal speed at latest fault	-	-	-	●
F2-17	Function selection at latest fault	-	-	-	●
F2-18	MCTC-PES-E1state at latest fault	-	-	-	●
F2-19	Input state 1 at latest fault	-	-	-	●
F2-20	Input state 2 at latest fault	-	-	-	●
F2-21	Output state at latest fault	-	-	-	●
F2-22	Phase A escalator speed at latest fault	-	-	-	●
F2-23	Phase B escalator speed at latest fault	-	-	-	●
F2-24	Detected running direction at latest fault	-	-	-	●
F2-25	Left handrail pulse time interval at latest fault	-	-	-	●
F2-26	Right handrail pulse time interval at latest fault	-	-	-	●

- F2-19 records the state of input terminals X1 to X16 when the latest fault occurs. Each terminal is allocated with a bit. The corresponding relationship is as follows:
Bit0: X1; Bit1: X2; Bit15: X16
- F2-20 records the state of input terminals X17 to X20 when the latest fault occurs. Each terminal is allocated with a bit. The corresponding relationship is as follows:
Bit0: X17 Bit1: X18; Bit2: X19; Bit3: X20
- F2-21 records the state of relays Y1 and Y3. Each relay is allocated with a bit. The corresponding relationship is as follows:
Bit0: Y1; Bit1: Y1F; Bit2: Y3; Bit3: Y3F
- F2-22 and F2-23 record the phase A/B escalator speed when the latest fault occurs. The actual record is phase A/B pulse cycle (unit: 10 ms) or the number of phase A/B pulses a second. They can be determined based on F2-10, F2-11 and F2-12.

Group FF: Factory Parameters

Group FP: Management Parameters

Function Code	Parameter Name	Setting Range	Default	Unit	Property
FP-00	User password	0-6553	0	-	★
FP-01	Software version	0-6553	-	-	●
FP-02	Reserved	-	-	-	●

4.4 Examples of Commissioning Applications

Here gives some commissioning application examples, in which the rated speed is set as the actual running speed.

- Suppose that escalator is used in the application. The rated speed of the escalator is 0.5 m/s, the radius of the driving chain sprocket is 0.3 m and the number of teeth of the driving chain sprocket is 72. Set parameters in group F0 as follows:

Function Code	Parameter Name	Setting Range	Setting
F0-00	System type	0: Escalator 1: Moving walkway	0
F0-01	Nominal speed	0.30-0.90	0.50
F0-02	Radius of driving chain sprocket	0.30-0.90	0.30
F0-03	PPR of driving chain sprocket	1-200	72
F0-04	Maximum braking-to-stop distance	1-200	Set according to China's standard

- 2) Generally, the step is 0.4 m wide. Set the radius of the handrail tachometer wheel to 0.05 m and there is only one inductive device on it. The following data can be obtained.

- F0-05: 0.63

Firstly calculate the perimeter of the handrail tachometer wheel: $0.05 \times 2 \times \pi \approx 0.314$. Then calculate the value of F0-05: $0.314/F0-01 = 0.63$.

- F0-06: 15.27

Firstly calculate the perimeter of the driving chain sprocket: $0.3 \times 2 \times \pi \approx 1.885$. Then calculate the value of F0-06: $0.4/1.885 \times 72 = 15.27$. Because F0-06 must be set a little larger than the calculation value (the automatic calculation value is integer 16), you can set F0-06 to any value in the range of 18 to 32.

- F0-07

You can set F0-07 to any value in the range of 0 to 14.

Note that the deviation between the setting of F0-06 and F0-07 and the automatic calculation result 16 must be a little large, at least by 2. For example, set F0-06 to 18 and F0-07 to 14.

Set parameters in group F0 as follows:

Function Code	Parameter Name	Setting Range	Setting
F0-05	Handrail pulse time interval at nominal speed	0.01–5.00	0.63
F0-06	Upper limit of phase A/B pulses for step signals	-	18
F0-07	Lower limit of phase A/B pulses for step signals	0 to F0-06	14

- 3) If the escalator does not have an auxiliary brake, set F0-08 to 2.

Function Code	Parameter Name	Setting Range	Setting
F0-08	Auxiliary function selection	0–65535	2



5

Troubleshooting

Chapter 5 Troubleshooting

The MCTC-PES-E1 system provides a total of 16 pieces of fault information and protective functions. It monitors various input signals, running conditions and external feedback information in real time. After a fault occurs, the system implements the protection function, and displays the fault code.

You can first determine the fault type, analyze the causes, and perform troubleshooting according to the following tables.

Table 5-1 Description of the faults of the MCTC-PES-E1

Fault Code	Fault Name	Description
ERR1	Overspeed of 1.2 times	The running speed exceeds 1.2 times of the nominal speed during normal running. If this fault occurs during commissioning, check whether the setting of parameters in group F0 is improper.
ERR2	Overspeed of 1.4 times	The running speed exceeds 1.4 times of the nominal speed during normal running. If this fault occurs during commissioning, check whether the setting of parameters in group F0 is improper.
ERR3	Unintentional reversal	Unintentional reversal occurs during the escalator running. If this fault occurs during commissioning, check whether the escalator speed detection signals (X15, X16) are reversed.
ERR4	Over-braking distance	The braking-to-stop distance exceeds the requirement. If this fault occurs during commissioning, check whether the setting of parameters in group F0 is improper.
ERR5	Left handrail underspeed	The left handrail speed is low. The setting of parameters in group F0 is improper. The sensor signal is abnormal.
ERR6	Right handrail underspeed	The right handrail speed is low. The setting of parameters in group F0 is improper. The sensor signal is abnormal.
ERR7	Up step loss	Check whether F0-06 is set to a value smaller than the actual value.
ERR8	Down step loss	Check whether F0-06 is set to a value smaller than the actual value.
ERR9	Service brake release fault	The working brake signal is abnormal.
ERR10	Auxiliary brake action fault	The mechanical switch feedback is inactive after braking. The auxiliary brake switch is active at startup. The auxiliary brake is not released at startup. The running in the upward direction exceeds 10s when the auxiliary brake switch is active. The auxiliary brake switch is active during running. The auxiliary brake contactor is disconnected during running.

Fault Code	Fault Name	Description
ERR11	Floor plate switch fault	The floor plate switch signal is active in normal state.
ERR12	External signal abnormal	1: There are A/B pulses in the stop state. 2: There is no A/B pulse within 4s after startup. 3: The number of A/B pulses for up step signals is smaller than the setting of F0-07. 4: The number of A/B pulses for down step signals is smaller than the setting of F0-07. 5: The left handrail pulse signals are very quick. 6: The right handrail pulse signals are very quick. 7: The two inspection signals are inconsistent. 8: The upward and downward signals are active simultaneously.
ERR13	MCTE-PES-E1 board hardware fault	1-4: The relay feedback is wrong. 5: The EEPROM initialization fails. 6: RAM check is wrong after power-on.
ERR14	EEPROM data wrong	-
ERR15	Main and auxiliary data check abnormal or MCU communication abnormal	The main and auxiliary chips are inconsistent in the following aspects: 1: Software versions of the main and auxiliary MCUs 2: States of the two chips 3: Signal state of terminals X1 to X4 4: Signal state of terminals X17 to X20 5: Output 6: Phase A escalator speed 7: Phase B escalator speed 9: Braking-to-stop distances detected by the main and auxiliary MCUs 8: The orthogonality of A and B pulses is not good and there is jump. 10: The left handrail signal is instable. 11: The right handrail signal is instable. 12, 13: The up step signal is instable. 14, 15: The down step signal is instable. 101-103: The communication between the main and the auxiliary chips is wrong. 104: Communication between the main and auxiliary chips fails at power-on.
ERR16	Parameter abnormal	101: The calculated number of pulses for 1.2 times of maximum braking-to-stop distance is wrong. 102: The calculated number of A/B pulses for step signals is wrong. 103: The calculated number of pulses a second is wrong.

Note

The number in the right column indicates the fault sub-code.

Table 5-2 System prompt for the faults of the MCTC-PES-E1

Fault Code	Fault Name	System Prompt
ERR1	Overspeed of 1.2 times	The LED blinks. The fault code output terminals (designated in Figure 3-9) output the fault code. If the operation panel is connected, it displays the fault code. The faults still exist after the system is powered on again.
ERR2	Overspeed of 1.4 times	
ERR3	Unintentional reversal	
ERR4	Over-braking distance	
ERR7/8	Up/Down step loss	
ERR9	Service brake release fault	The system prompts are the same as above, but the faults are reset after the system is re-powered on.
ERR5/6	Left/Right handrail underspeed	
ERR10	Auxiliary brake action fault	
ERR12/13/14/15	Signal abnormal or MCTC-PES-E1 faulty	The system prompts are the same as above, but the faults are reset after the system is powered on again.
ERR11	Floor plate switch fault	

Warranty Agreement

1. The warranty period of the product is 18 months (refer to the barcode on the equipment). During the warranty period, if the product fails or is damaged under the condition of normal use by following the instructions, Monarch will be responsible for free maintenance.
2. Within the warranty period, maintenance will be charged for the damages caused by the following reasons:
 - a. Improper use or repair/modification without prior permission
 - b. Fire, flood, abnormal voltage, other disasters and secondary disaster
 - c. Hardware damage caused by dropping or transportation after procurement
 - d. Improper operation
 - e. Damage out of the equipment (for example, external device)
3. If there is any failure or damage to the product, please correctly fill out the Product Warranty Card in detail.
4. The maintenance fee is charged according to the latest Maintenance Price List of Monarch.
5. The Product Warranty Card is not re-issued. Please keep the card and present it to the maintenance personnel when asking for maintenance.
6. If there is any problem during the service, contact Monarch's agent or Monarch directly.
7. This agreement shall be interpreted by Suzhou MONARCH Control Technology Co., Ltd.

Service Department, Suzhou MONARCH Control Technology Co., Ltd.

Address: 16, Youciang Rd, Wangshan Industrial Park, Wuzhong Economic Development Zone, Suzhou, China P.C.: 215104

Website: <http://www.szmctc.cn>

Aonarch

Product Warranty Card

Customer information	Add. of unit:	
	Name of unit: P.C.:	Contact person:
		Tel.:
Product information	Product model:	
	Body barcode (Attach here):	
	Name of agent:	
Failure information	(Maintenance time and content):	
	Maintenance personnel:	